

GPV Preimage Sampling with Weak Smoothness and Its Applications to Lattice Signatures

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Abstract. The lattice trapdoor associated with Ajtai’s function is the cornerstone of many lattice-based cryptosystems. The current provably secure trapdoor framework, known as the GPV framework, uses a *strong smoothness* condition, i.e. $\epsilon \ll \frac{1}{n^2}$ for smoothing parameter $\eta_\epsilon(\mathbb{Z}^n)$, to ensure the correctness of the security reduction.

In this work, we investigate the feasibility of *weak smoothness*, e.g. $\epsilon = O(\frac{1}{n})$ or even $O(1)$ in the GPV framework and present several positive results. First, we provide a theoretical security proof for GPV with weak smoothness under a new assumption. Then, we present Gaussian samplers that are compatible with the weak smoothness condition. As direct applications, we present several practical GPV signature instantiations based on a weak smoothness condition. Our first instantiation is a variant of FALCON, called FALCON^{ws}, achieving *smaller size* and *higher security*. The public key sizes are 21% to 28% smaller, and the signature sizes are 23.5% to 29% smaller than FALCON. We also showcase an NTRU-based GPV signature scheme that employs the Peikert sampler with weak smoothness. This offers a simple implementation while the security level is greatly lower. Nevertheless, at the NIST-3 security level, our scheme achieves a 49% reduction in size compared to Dilithium-3. We also derive a weak smoothness variant of the ANTRAG signature scheme, called ANTRAG^{ws}, along with a floating-point free version, which offers an attractive trade-off among portability, efficiency and security. Compared to ANTRAG, ANTRAG^{ws} achieves reductions of 16.5% to 21.8% in signature size and 14.3% to 21.4% in public key size. Furthermore, we adapt the compact gadget framework to the weak smoothness setting, removing

the need for the new assumption and improving concrete parameters of gadget-based signatures, e.g. HUFU.

Keywords: Lattice-based cryptography · GPV trapdoor · Gaussian sampling · FALCON Signature Scheme

1 Introduction

Lattice-based cryptography is a primary and promising area of post-quantum cryptography. In 2022, the US NIST announced four selected algorithms slated for post-quantum cryptography standardization, three of which are lattice-based: Kyber [PS22], Dilithium [LDK⁺22], and Falcon [PFH⁺22]. Lattice-based cryptography also offers a wide variety of applications, such as fully homomorphic encryption [Gen09], anonymous credentials [BLNS23, JRLS23], attribute-based encryption [GVW15], and much more [Pei16, PS19, EZS⁺19].

Many lattice-based cryptosystems are built from lattice trapdoors associated with Ajtai’s function $f_{\mathbf{A}}(\mathbf{x}) = \mathbf{Ax} \bmod q$, where $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ is a wide matrix. For random $\mathbf{A}$, finding a short preimage $\mathbf{x}$ of a given syndrome $\mathbf{u}$, i.e., $\mathbf{u} = f_{\mathbf{A}}(\mathbf{x})$, is as hard as solving certain worst-case lattice problems [Ajt96]. However, with a trapdoor that is equivalent to a short basis of the q -ary lattice $\{\mathbf{x} \in \mathbb{Z}^m : \mathbf{Ax} = 0 \bmod q\}$, one can efficiently compute a short preimage. An important application of lattice trapdoors is digital signatures, in which the syndrome is computed as the hashed message, and the short preimage serves as the signature. Lattice signatures generally require particular care in preventing secret leaks from signature transcripts. Some early proposals were broken by statistical attacks exploiting such leaks [NR06, DN12, YD18, LSZ⁺24].

In 2008, Gentry, Peikert, and Vaikuntanathan proposed a provably secure lattice trapdoor framework [GPV08], known as the GPV framework. At the heart of the GPV schemes is a polynomial-time algorithm sampling preimages from some lattice Gaussian. Since a lattice Gaussian, i.e., the preimage distribution, is independent of the lattice trapdoor, this allows us to simulate the sampling procedure without knowing the secret trapdoor, which immediately gives a security proof in the random oracle model. Currently, there are two main paradigms for lattice Gaussian sampling: the Klein-GPV sampler [Kle00, GPV08] and the Peikert sampler [Pei10].

The Gaussian width s is a crucial parameter for the GPV signatures. The smaller s is, the shorter the preimages are, which also implies a smaller signature size and higher security against forgery attacks. Typically, the minimal Gaussian width achieved can be written as

$$s_{\min} = Q(\mathbf{T}) \cdot \eta_{\epsilon}(\mathbb{Z}^m)$$

where $Q(\mathbf{T})$ denotes the quality of the trapdoor $\mathbf{T}$ and $\eta_{\epsilon}(\mathbb{Z}^m)$ is the smoothing parameter of $\mathbb{Z}^m$ [MR07]. The trapdoor quality is related to the trapdoor itself and the used Gaussian sampler. The parameter ϵ in $\eta_{\epsilon}(\mathbb{Z}^m)$ is used to evaluate the

closeness between the distribution output by the sampler and the ideal Gaussian. With ϵ decreasing, two distributions get increasingly close but $\eta_\epsilon(\mathbb{Z}^m)$ becomes larger resulting in a larger Gaussian width.

To improve the practicality of GPV schemes, several prior works were proposed to explore trapdoor constructions achieving smaller $Q(\mathbf{T})$. Nowadays, practical GPV trapdoors can be basically classified into two families: NTRU trapdoor and gadget trapdoor. The gadget trapdoor was first proposed in [MP12] and turned out to be particularly convenient for advanced cryptographic constructions. However, for basic signature applications, gadget trapdoors are still less efficient than NTRU trapdoors, even though recent techniques have greatly improved the performance of gadget trapdoor schemes [CGM19, YJW23]. Practical NTRU trapdoor-based GPV signatures date back to the work of Ducas, Lyubashevsky, and Prest [DLP14]. They discovered that NTRU can admit a compact trapdoor achieving nearly optimal quality, and based on this, they proposed a first practical GPV signature scheme. Later, FALCON [PFH⁺22] further improved this construction by integrating a ring-efficient Klein-GPV sampler from the fast Fourier orthogonalization [DP16]. FALCON offers good performance both in terms of speed and size, however, its signing algorithm is highly complicated. To simplify the signing procedure of FALCON, Espitau et al. designed the MITAKA signature scheme [EFG⁺22] based on the hybrid sampler [DP15]. Compared to the Klein-GPV sampler, the hybrid sampler works with a larger Gaussian width, which reduces the security. To mitigate the security loss, MITAKA uses some techniques to refine the NTRU trapdoor quality. A follow-up work [ENS⁺23] presents an improved trapdoor generation algorithm and finally makes MITAKA as secure as FALCON.

In addition to the aforementioned works on optimizing the trapdoor quality $Q(\mathbf{T})$, there are also a batch of works studying how small the term $\eta_\epsilon(\mathbb{Z}^m)$ can be. In the original GPV framework, ϵ is set as $2^{-\lambda}$ for λ -bits of security. By using Kullback-Leibler divergence to replace statistical distance as the closeness measurement, [DLP14] showed that ϵ can be relaxed to $2^{-\lambda/2}$. Based on Rényi divergence and taking the number of signature queries Q_s into account, Prest suggested a tighter bound [Pre17]: $\epsilon \leq \frac{1}{\sqrt{Q_s \cdot \lambda}}$, which was taken by Falcon. For practical signatures, it is routine to set $Q_s = 2^{64}$ and $n \in (500, 2000)$, hence $\epsilon \leq \frac{1}{\sqrt{Q_s \cdot \lambda}} \ll \frac{1}{n^2}$. We term this a *strong smoothness* condition.

Our Contributions. In this work, we investigate the feasibility of *weak smoothness*, e.g. $\epsilon = O(\frac{1}{n})$ even $O(1)$ in the GPV framework and give some positive results. Our contributions are mainly threefold.

First, we provide a theoretical security proof for GPV with weak smoothness. For a relatively larger ϵ , solely by smoothing parameter property, the previous security proof of strong unforgeability does not hold. The issue is that the distribution of simulated syndrome by the adversary is no longer statistically indistinguishable from uniform, thus, the adversary cannot simply program the random oracle.

To this end, we introduce a new assumption called χ -BDD $^{\mathcal{O}_s}$ to guarantee the simulated syndrome is *computationally indistinguishable* from uniform. By this, the adversary still cannot distinguish the real signature distribution and the simulated one, and the proof follows. We conduct some cryptanalysis on the new assumption and show that it has no impact on the concrete security of practical GPV signatures.

Second, we present Gaussian samplers compatible with the weak smoothness condition. Under weak smoothness, both Klein-GPV and Peikert samplers cannot output a distribution sufficiently close to the target Gaussian, which possibly leaks secret information. To this end, we apply the rejection sampling to rectify the distribution. In fact, this idea was first suggested in [BLP⁺13] and a modified Klein-GPV sampler was proposed to implement *exact* Gaussian sampling. We extend this idea to Peikert sampler and some extra treatments are used. We further adapt the hybrid sampler [Pre15] in the weak smoothness setting and provide a refined analysis of the expected number of repetitions. In addition, we describe an efficient algorithm to compute the ratio of two Gaussian masses $\rho_{\sigma,c}(\mathbb{Z})$ and $\rho_{\sigma}(\mathbb{Z})$. Compared to the Poisson summation based method [BLP⁺13], our algorithm makes use of the product representation of the theta function, and we show that the resulting approximation converges rapidly.

Our third contribution is to give several practical GPV signature schemes using the weak smoothness condition. We first present a variant of FALCON, denoted FALCON^{ws}, achieving *smaller size* and *higher security*. We set $\epsilon = \frac{1}{2n}$ and experimentally verified that such a large ϵ would only lead to a moderate efficiency loss: the average rejection number is less than 1.3 for $n = 512, 1024$. More interestingly, FALCON^{ws} allows us to reduce the modulus q while keeping the signature norm bound less than q . In fact, simply reducing q as in [ETWY22] would make the signature norm bound exceed q , which opens a door to the Z-shape attack [DEP23]. Table 1 shows the comparisons between FALCON and FALCON^{ws}.

Table 1. Comparisons between FALCON and FALCON^{ws}. The numbers for public key and signature sizes are in bytes. C (resp. Q) represents the bit security in classical (resp. quantum) setting.

	ϵ of $\eta_{\epsilon}(\mathbb{Z})$	Modulus q	PK Size	Sig. Size	Security (C/Q)
FALCON-512	$2^{-45.5}$	12289	897	666	120/108
FALCON ^{ws} -512	2^{-10}	953	640	474	142/129
FALCON-1024	2^{-47}	12289	1793	1280	273/248
FALCON ^{ws} -1024	2^{-11}	1949	1408	979	284/258

We next leverage the hybrid sampler under weak smoothness condition to derive a variant of ANTRAG [ENS⁺23], called ANTRAG^{ws}. Choosing $\epsilon = \frac{1}{8n}$,

ANTRAG^{ws} can safely work with a smaller modulus while keeping the expected repetitions below 1.28 for $n = 512, 1024$. Table 2 presents a detailed comparison between ANTRAG and ANTRAG^{ws}. Moreover, we propose a variant of ANTRAG^{ws} which does not need floating-point arithmetic, offering better portability and avoiding potential security risk [LTYZ25]. Unlike the original MITAKAZ_ℤ [EFG⁺22], we sample the perturbation with an approximate decomposition (of precision $1/p$) of the covariance matrix and uses another rejection sampling to correct the discrepancy caused by rounding errors. The acceptance probability remains close to one for some moderate p , resulting in negligible efficiency loss.

Table 2. Comparisons between ANTRAG and ANTRAG^{ws}. The numbers for public key and signature sizes are in bytes.

	ϵ of $\eta_\epsilon(\mathbb{Z})$	Modulus q	PK Size	Sig. Size	Security (C/Q)
ANTRAG-512	$2^{-45.5}$	12289	896	646	124/113
ANTRAG ^{ws} -512	2^{-12}	1237	704	505	131/119
ANTRAG-1024	2^{-47}	12289	1792	1260	264/240
ANTRAG ^{ws} -1024	2^{-13}	2729	1536	1052	270/245

We also showcase an NTRU-based GPV signature scheme using Peikert sampler with weak smoothness. This offers a simpler implementation than Mitaka [EFG⁺22] while the security level is greatly lower. Nevertheless for the security level NIST-3, our scheme is 49% smaller than Dilithium-3. Table 3 shows the concrete numbers.

Table 3. Comparisons between Dilithium-3 and our GPV signature scheme using a modified Peikert sampler. The numbers for public key and signature sizes are in bytes.

	PK Size	Sig. Size	Security (C/Q)
Dilithium-3	1952	3293	182/165
Our scheme	2176	1655	189/171

Finally, we adapt the compact gadget framework [YJW23] to the weak smoothness setting. Since the provable security of this framework relies on trapdoor sampling in an average-case sense, we deal with the weak smoothness condition in a manner different from that in the case of aforementioned NTRU trapdoor based signatures. In particular, compact gadgets under weak smoothness no longer require an additional χ -BDD ^{$\mathcal{O}_s$} assumption. We further give a variant of the HuFU signature algorithm [YJL⁺23] under weak smoothness condition.

Compared to the original HuFu scheme, HuFu^{ws} achieves a reduction of $> 13\%$ in signature size and $> 17\%$ in public key size (see Table 4 for details).

Table 4. Comparisons between HuFu and HuFu^{ws} . The number for public key size is in kilobytes and that for signature size is in bytes.

	Scheme	Modulus	PK Size	Sig. Size
NIST-1	HuFu	2^{16}	1059	2455
	HuFu^{ws}	2^{15}	807	2079
NIST-3	HuFu	2^{17}	2177	3540
	HuFu^{ws}	2^{16}	1807	3071
NIST-5	HuFu	2^{17}	3573	4520
	HuFu^{ws}	2^{16}	2927	3915

As a side contribution, we present a modified rANS encoding to prevent the implementation attack against HAETAE [CCD⁺23] and HuFu [YJL⁺23] signature schemes presented by Markku-Juhani O. Saarinen [Saa23b,Saa23a] while keeping a good compression efficiency.

Differences from the Conference Version. The previous version of this paper appeared in Asiacrypt 2025 [ZJR⁺25]. The main differences between this version and the conference version are as follows:

- In Section 4.3, we extend the hybrid sampler to the weak smoothness condition, which solves the problem left in the conference version. The adaption and the proof are simple however embarrassingly not realized before.
- In Section 4.4, we theoretically prove the convergence rate of the Jacobi triple-product based Theta function division (Algorithm 5), which supports the experimental results presented in the conference version.
- In Section 5.3, we present yet another NTRU-based signature $\text{ANTRAG}^{\text{ws}}$, a weak smoothness variant of ANTRAG, with well-chosen parameters. For better portability, we also give a variant of $\text{ANTRAG}^{\text{ws}}$ not relying on floating-point arithmetic.
- In Section 6, we apply the weak smoothness condition to the compact gadget framework and improve the size of gadget-based signatures. This partially resolves an open problem in the conference version.

Roadmap. Following some preliminary material in Section 2, we present a security proof for GPV signatures with a weak smoothness condition in Section 3. Section 4 adapts the Klein-GPV sampler, the Peikert sampler and the hybrid sampler in the weak smoothness setting. We instantiate several practical NTRU trapdoor based signature schemes with weak smoothness in Section 5

and formalize compact gadget trapdoors under the weak smoothness condition in Section 6. Section 7 gives a secure rANS encoding for signature compression. The source codes of the experiments in this paper are available at https://github.com/zsdthu/Weak_Smoothness.

2 Preliminaries

We use bold lowercase (resp. uppercase) letters for vectors (resp. matrices). By convention, vectors are in column form. For a distribution D , we write $z \leftarrow D$ if the random variable z is sampled from D and denote by $D(x)$ the probability that $z = x$. We denote by $z \sim D$ a random variable distributed as D . For a real-valued function f and a countable set S , we write $f(S) = \sum_{x \in S} f(x)$ assuming this sum is absolutely convergent. We use the notational shortcut $\hat{\epsilon} = \epsilon + O(\epsilon^2)$. We use the convention $\mathbb{N}^+ = \{1, 2, 3, \dots\}$.

2.1 Linear Algebra and Lattices

Let b_i (resp. $\mathbf{b}_i$) denote the i -th coordinate (resp. column) of $\mathbf{b}$ (resp. $\mathbf{B}$). Given $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, their inner product is $\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{i=1}^n u_i v_i$. When $\langle \mathbf{u}, \mathbf{v} \rangle = 0$, we call $\mathbf{u}$ and $\mathbf{v}$ are orthogonal. Let $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$ be the ℓ_2 -norm of $\mathbf{v}$. Let $\mathbf{I}_n$ denote the n -dimensional identity matrix. Let $\mathbf{B}^t$ be the transpose of $\mathbf{B}$.

We write $\Sigma > 0$ (resp., $\Sigma \geq 0$) when a symmetric matrix $\Sigma \in \mathbb{R}^{n \times n}$ is *positive definite* (resp. *semidefinite*), i.e. $\mathbf{x}^t \Sigma \mathbf{x} > 0$ (resp., $\mathbf{x}^t \Sigma \mathbf{x} \geq 0$) for all nonzero $\mathbf{x} \in \mathbb{R}^n$. We write $\Sigma_1 \geq \Sigma_2$ or $\Sigma_2 \leq \Sigma_1$ if $\Sigma_1 - \Sigma_2 \geq 0$, and similarly for $\Sigma_1 > \Sigma_2$. It holds that $\Sigma_1 > \Sigma_2 > 0$ if and only if $\Sigma_2^{-1} > \Sigma_1^{-1} > 0$. If $\Sigma = \mathbf{A} \mathbf{A}^t$, we call $\mathbf{A}$ a *Gram root* of Σ . Let $\sqrt{\Sigma}$ denote any Gram root of Σ when the context permits it.

For a matrix $\mathbf{A} \in \mathbb{R}^{d \times m}$, there exists a unique matrix $\mathbf{A}^+$ verifying the four Moore-Penrose conditions: (1) $\mathbf{A} \mathbf{A}^+ \mathbf{A} = \mathbf{A}$, (2) $\mathbf{A}^+ \mathbf{A} \mathbf{A}^+ = \mathbf{A}^+$, (3) $(\mathbf{A} \mathbf{A}^+)^t = \mathbf{A} \mathbf{A}^+$, (4) $(\mathbf{A}^+ \mathbf{A})^t = \mathbf{A}^+ \mathbf{A}$. The matrix $\mathbf{A}^+$ is called the Moore-Penrose pseudoinverse, or just pseudoinverse for short, of $\mathbf{A}$. If $\mathbf{A}$ has linearly independent columns, then $\mathbf{A}^+ = (\mathbf{A}^t \mathbf{A})^{-1} \mathbf{A}^t$.

Let $\mathbf{B} = (\mathbf{b}_0, \dots, \mathbf{b}_{n-1}) \in \mathbb{R}^{m \times n}$ be a basis of rank n . The Gram-Schmidt Orthogonalization (GSO) of $\mathbf{B}$ is the unique matrix $\tilde{\mathbf{B}} = (\tilde{\mathbf{b}}_0, \dots, \tilde{\mathbf{b}}_{n-1}) \in \mathbb{R}^{m \times n}$ such that $\mathbf{B} = \tilde{\mathbf{B}} \mathbf{U}$ where $\tilde{\mathbf{b}}_i$'s are pairwise orthogonal and $\mathbf{U}$ is upper-triangular with 1 on its diagonal. Let $\|\mathbf{B}\|_{GS} = \max_i \|\tilde{\mathbf{b}}_i\|$. The largest singular value of $\mathbf{B}$ is denoted by $s_1(\mathbf{B}) = \max_{\mathbf{x} \neq \mathbf{0}} \frac{\|\mathbf{B} \mathbf{x}\|}{\|\mathbf{x}\|}$.

A lattice $\mathcal{L}$ is the set of all integer linear combinations of linearly independent vectors $\mathbf{b}_0, \dots, \mathbf{b}_{n-1} \in \mathbb{R}^m$, i.e. $\mathcal{L} = \left\{ \sum_{i=0}^{n-1} x_i \mathbf{b}_i \mid x_i \in \mathbb{Z} \right\}$. We call $\mathbf{B} = (\mathbf{b}_0, \dots, \mathbf{b}_{n-1})$ a basis and n the dimension of $\mathcal{L}$. Let $\mathcal{L}(\mathbf{B})$ denote the lattice generated by a basis $\mathbf{B}$. The dual lattice of $\mathcal{L}$ is defined as $\mathcal{L}^* = \{\mathbf{w} \in \mathbb{R}^n \mid \langle \mathbf{x}, \mathbf{w} \rangle \in \mathbb{Z}, \mathbf{x} \in \mathcal{L}\}$. Many cryptographic applications rely on a specific family of lattices known as q -ary lattice. Given $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$, let $\mathcal{L}^\perp(\mathbf{A}) = \{\mathbf{x} \in \mathbb{Z}^m :$

$\mathbf{Ax} = \mathbf{0} \bmod q\}$. Each $\mathbf{u} \in \mathbb{Z}_q^n$ defines a lattice coset $\mathcal{L}_{\mathbf{u}}^\perp(\mathbf{A}) = \{\mathbf{x} \in \mathbb{Z}^m : \mathbf{Ax} = \mathbf{u} \bmod q\}$.

Let us recall two lattice hard problems. The SIS problem is in the so-called Hermite Normal Form. The χ -BDD $_{\mathcal{L}}$ problem is formalized and studied in [DP23].

Definition 1 (SIS). Let $\mathbf{A}$ be a uniformly random matrix over $\mathbb{Z}_q^{n \times m}$ and $\beta > 0$. The SIS problem asks to find a non-zero integer vector $(\mathbf{x}_0, \mathbf{x}_1)$ such that $[\mathbf{I}_n \ \mathbf{A}] \begin{bmatrix} \mathbf{x}_0 \\ \mathbf{x}_1 \end{bmatrix} = \mathbf{0} \bmod q$ and $\|(\mathbf{x}_0, \mathbf{x}_1)\| < \beta$.

Definition 2 (χ -BDD $_{\mathcal{L}}$). Let $\chi : \mathbb{Z}^n \rightarrow [0, 1]$ be some distribution and $\mathcal{L} \subseteq \mathbb{Z}^n$ be a full-rank lattice. The χ -BDD $_{\mathcal{L}}$ problem asks to determine (with high probability) whether an input $\mathbf{t}$ is sampled from $\chi \bmod \mathcal{L}$ or $U(\mathbb{Z}^n / \mathcal{L})$.

2.2 Gaussian Distributions

The n -dimensional *Gaussian* function $\rho : \mathbb{R}^n \rightarrow (0, 1]$ is defined as $\rho(\mathbf{x}) := \exp(-\pi \|\mathbf{x}\|^2)$. Let $\rho_{\mathbf{B}}(\mathbf{x}) = \exp(-\pi \mathbf{x}^t \mathbf{\Sigma}^{-1} \mathbf{x})$ where $\mathbf{\Sigma} = \mathbf{BB}^t$. Since $\rho_{\mathbf{B}}(\mathbf{x})$ is completely determined by $\mathbf{\Sigma} = \mathbf{BB}^t$, we also write $\rho_{\sqrt{\mathbf{\Sigma}}}(\mathbf{x}) = \rho_{\mathbf{B}}(\mathbf{x})$. Let $\rho_{\sqrt{\mathbf{\Sigma}}, \mathbf{c}}(\mathbf{x}) = \rho_{\sqrt{\mathbf{\Sigma}}}(\mathbf{x} - \mathbf{c})$ for $\mathbf{c} \in \text{span}(\mathbf{\Sigma})$. When $\mathbf{c} = \mathbf{0}$, we omit the subscript $\mathbf{c}$. The normal distribution $\mathcal{N}_{\sqrt{\mathbf{\Sigma}}}$ of (scaled) covariance $\mathbf{\Sigma}$ then has density probability function $\det(\mathbf{\Sigma})^{-\frac{1}{2}} \rho_{\sqrt{\mathbf{\Sigma}}}$. For a countable set $S \subset \mathbb{R}^n$, let $\rho_{\sqrt{\mathbf{\Sigma}}}(S) = \sum_{\mathbf{s} \in S} \rho_{\sqrt{\mathbf{\Sigma}}}(\mathbf{s})$. The *discrete Gaussian* over a lattice $\mathcal{L}$ with center $\mathbf{c}$ and covariance matrix $\mathbf{\Sigma}$ is defined by the probability function

$$D_{\mathcal{L}, \sqrt{\mathbf{\Sigma}}, \mathbf{c}}(\mathbf{x}) = \frac{\rho_{\sqrt{\mathbf{\Sigma}}, \mathbf{c}}(\mathbf{x})}{\rho_{\sqrt{\mathbf{\Sigma}}, \mathbf{c}}(\mathcal{L})} \propto \rho_{\sqrt{\mathbf{\Sigma}}, \mathbf{c}}(\mathbf{x}).$$

The discrete Gaussian on $\mathcal{L} + \mathbf{c}$, for $\mathbf{c} \in \text{span}(\mathcal{L})$, is defined by $D_{\mathcal{L} + \mathbf{c}, \sqrt{\mathbf{\Sigma}}}(\mathbf{x}) = \rho_{\sqrt{\mathbf{\Sigma}}}(\mathbf{x}) / \rho_{\sqrt{\mathbf{\Sigma}}}(\mathcal{L} + \mathbf{c})$ for all $\mathbf{x} \in \mathcal{L} + \mathbf{c}$. When $\mathbf{\Sigma} = s^2 \mathbf{I}$, we call the Gaussian *spherical* of *width* s and write the subscript $\sqrt{\mathbf{\Sigma}}$ as s simply. The width s play the same role as the standard deviation σ . Actually, $s = \sqrt{2\pi} \sigma$. In this paper, the symbol σ is used only to represent the standard deviation. For $c \in \mathbb{R}, r > 0$, we note $[c]_r$ the distribution $D_{\mathbb{Z}, c, r}$, and this notation is generalized coefficient-wise when c is replaced with a vector $\mathbf{c}$.

For a lattice $\mathcal{L}$ and $\epsilon > 0$, $\eta_\epsilon(\mathcal{L}) = \min\{s > 0 \mid \rho_{\frac{1}{s}}(\mathcal{L}^*) \leq 1 + \epsilon\}$ is called the *smoothing parameter*. The following definition is a generalized version.

Definition 3 ([Pei10, Definition 2.3]). Let $\mathbf{\Sigma} > 0$ and lattice $\mathcal{L} \in \text{span}(\mathbf{\Sigma})$. We write $\sqrt{\mathbf{\Sigma}} \geq \eta_\epsilon(\mathcal{L})$ if $\eta_\epsilon(\sqrt{\mathbf{\Sigma}}^{-1} \cdot \mathcal{L}) \leq 1$ i.e. $\rho_{\sqrt{\mathbf{\Sigma}^{-1}}}(\mathcal{L}^*) \leq 1 + \epsilon$.

Notice that for two lattices of the same rank $\mathcal{L}_1 \subseteq \mathcal{L}_2$, the denser lattice always has the smaller smoothing parameter, i.e. $\eta_\epsilon(\mathcal{L}_2) \leq \eta_\epsilon(\mathcal{L}_1)$. Let $\bar{\eta}_\epsilon(\mathbb{Z}) = \sqrt{\frac{\ln(2(1+1/\epsilon))}{\pi}}$. Here we recall several facts to be used later.

Lemma 1 ([MR07, implicit in Lemma 4.4]). For $\epsilon \in (0, 1)$,

1. $\eta_\epsilon(\mathbb{Z}) \leq \overline{\eta}_\epsilon(\mathbb{Z})$
2. if $s \geq \eta_\epsilon(\mathbb{Z})$, then $\rho_{s,c}(\mathbb{Z}) \in \left[\frac{1-\epsilon}{1+\epsilon}, 1\right] \cdot \rho_s(\mathbb{Z})$.

Lemma 2 ([GPV08, Lemma 5.2]). Assume the columns of $\mathbf{A} \in \mathbb{Z}^{n \times m}$ generate $\mathbb{Z}^n$, and $s \geq \eta_\epsilon(\mathcal{L}^\perp(\mathbf{A}))$. Then for $\mathbf{e} \leftarrow D_{\mathbb{Z}^m, s}$, the distribution of the syndrome $\mathbf{u} = \mathbf{A}\mathbf{e} \bmod q$ is within statistical distance 2ϵ of uniform over $\mathbb{Z}_q^n$.

Lemma 3 ([Pei10, Fact 2.1]). Let $\Sigma_1, \Sigma_2 \in \mathbb{R}^{n \times n}$ be positive definite matrices and $\mathbf{x}, \mathbf{c}_1, \mathbf{c}_2 \in \mathbb{R}^n$. We have

$$\rho_{\sqrt{\Sigma_1}}(\mathbf{x} - \mathbf{c}_1) \rho_{\sqrt{\Sigma_2}}(\mathbf{x} - \mathbf{c}_2) = \rho_{\sqrt{\Sigma_0}}(\mathbf{c}_1 - \mathbf{c}_2) \rho_{\sqrt{\Sigma_3}}(\mathbf{x} - \mathbf{c}')$$

where $\Sigma_0 = \Sigma_1 + \Sigma_2$, $\Sigma_3^{-1} = \Sigma_1^{-1} + \Sigma_2^{-1}$ and $\Sigma_3^{-1}\mathbf{c}' = \Sigma_1\mathbf{c}_1 + \Sigma_2\mathbf{c}_2$.

Theorem 1 (Adapted from [Pei10, Theorem 3.1]). Let $\Sigma_1, \Sigma_2 \in \mathbb{R}^{n \times n}$ be positive definite matrices. Let $\Sigma = \Sigma_1 + \Sigma_2$ and let $\Sigma_3 \in \mathbb{R}^{n \times n}$ be such that $\Sigma_3^{-1} = \Sigma_1^{-1} + \Sigma_2^{-1}$. Let $\mathcal{L}_1, \mathcal{L}_2$ be two full-rank lattices in $\mathbb{R}^n$ such that $\sqrt{\Sigma_1} \geq \eta_\epsilon(\mathcal{L}_1)$ and $\sqrt{\Sigma_3} \geq \eta_\epsilon(\mathcal{L}_2)$ for $\epsilon \in (0, 1/2)$. Let $\mathbf{c}_1, \mathbf{c}_2 \in \mathbb{R}^n$, then the distribution of $\mathbf{x} \leftarrow D_{\mathcal{L}_1, \sqrt{\Sigma_1}, \mathbf{p} - \mathbf{c}_2 + \mathbf{c}_1}$ where $\mathbf{p} \leftarrow D_{\mathcal{L}_2, \sqrt{\Sigma_2}, \mathbf{c}_2}$ is within max-log distance $4\hat{\epsilon}$ of $D_{\mathcal{L}_1, \sqrt{\Sigma}, \mathbf{c}_1}$.

Lemma 4 ([Dad14, Theorem 12]). Let $\mathcal{L} \subset \mathbb{R}^n$ be a full-rank lattice, $\mathbf{t} \in \mathbb{R}^n$ and $X \sim D_{\mathcal{L} + \mathbf{t}, s}$. Then, for $r \geq 1$, we have

$$\Pr \left[\|X\| > rs \sqrt{\frac{n}{2\pi}} \right] \leq \frac{\rho_s(\mathcal{L})}{\rho_s(\mathcal{L} + \mathbf{t})} \exp\left(-\frac{n}{2}(r-1)^2\right).$$

2.3 The Rényi Divergence

We use the same definition of the Rényi divergence as in [BLRL⁺18, Pre17].

Definition 4. Let $\mathcal{P}, \mathcal{Q}$ be two distributions such that $\text{Supp}(\mathcal{P}) \subseteq \text{Supp}(\mathcal{Q})$. For $a \in (1, +\infty)$, we define the Rényi divergence of order a by

$$R_a(\mathcal{P} \parallel \mathcal{Q}) = \left(\sum_{x \in \text{Supp}(\mathcal{P})} \frac{\mathcal{P}(x)^a}{\mathcal{Q}(x)^{a-1}} \right)^{\frac{1}{a-1}}$$

. In addition, the Rényi Divergence of order $+\infty$ is defined as

$$R_\infty(\mathcal{P} \parallel \mathcal{Q}) = \max_{x \in \text{Supp}(\mathcal{P})} \frac{\mathcal{P}(x)}{\mathcal{Q}(x)}.$$

Generic (resp. cryptographic) properties of the Rényi divergence can be found in [BLRL⁺18]. We recall the most important ones.

Lemma 5 ([BLRL⁺18, Lemma 2.9]). *For two distributions $\mathcal{P}, \mathcal{Q}$ and two families of distributions $(\mathcal{P}_i)_i, (\mathcal{Q}_i)_i$, the Rényi divergence verifies the following properties:*

- Data processing inequality. *For any function f , $R_a(\mathcal{P}^f \| \mathcal{Q}^f) \leq R_a(\mathcal{P} \| \mathcal{Q})$.*
- Multiplicativity. *$R_a(\prod_i \mathcal{P}_i \| \prod_i \mathcal{Q}_i) = \prod_i R_a(\mathcal{P}_i \| \mathcal{Q}_i)$.*
- Weak Triangle Inequality. *$R_a(\mathcal{P}_1 \| \mathcal{P}_3) \leq R_a(\mathcal{P}_1 \| \mathcal{P}_2) \cdot R_\infty(\mathcal{P}_2 \| \mathcal{P}_3)$.*

Lemma 6 ([Pre17, Lemma 2]). *Let D and D_δ be two distributions such that:*

- $\exists \delta > 0$ such that $\frac{D_\delta}{D} \leq 1 + \delta$ over $\text{Supp}(D_\delta)$

Then for $a \in (1, +\infty]$:

$$R_a(D_\delta \| D) \leq 1 + \delta.$$

2.4 Power-of-two Cyclotomics and NTRU

Let $n = 2^\ell$ for some integer $\ell \geq 1$ and ζ_n to be a $2n$ -th primitive root of 1. For a fixed n , $\mathcal{K} := \mathbb{Q}(\zeta_n)$ is the n -th power-of-two cyclotomic field, and its ring of algebraic integers is $\mathcal{R} := \mathbb{Z}[\zeta_n]$. The field automorphism $\zeta_n \mapsto \zeta_n^{-1}$ corresponds to the complex conjugation, and we write the image f^* of f under this automorphism. We have $\mathcal{K} \simeq \mathbb{Q}[x]/(x^n + 1)$ and $\mathcal{R} \simeq \mathbb{Z}[x]/(x^n + 1)$, and both are contained in $\mathcal{K}_\mathbb{R} = \mathcal{K} \otimes \mathbb{R} \simeq \mathbb{R}[x]/(x^n + 1)$. Each $f = \sum_{i=0}^{n-1} f_i \zeta_n^i \in \mathcal{K}_\mathbb{R}$ can be identified with its coefficient vector $(f_0, \dots, f_{n-1}) \in \mathbb{R}^d$. The cyclotomic field $\mathcal{K}$ comes with n complex field emmbedding $\varphi_i : \mathcal{K} \rightarrow \mathbb{C}$ which map f seen as a polynomial to its evaluations at the odd powers of ζ_n . This defines the so-called canonical embedding $\varphi(f) := (\varphi_1(f), \dots, \varphi_n(f))$. It extends straightforwardly to $\mathcal{K}_\mathbb{R}$. Note that $\varphi(fg) = (\varphi_i(f)\varphi_i(g))_{i \leq n}$.

There is a natural Hermitian form on $\mathcal{K}^m$ defined by $\langle \mathbf{x}, \mathbf{y} \rangle_\mathcal{K} := \sum_{i=0}^{m-1} x_i^* y_i \in \mathcal{K}$ where $\mathbf{x} = (x_0, \dots, x_{m-1})$, $\mathbf{y} = (y_0, \dots, y_{m-1})$ span $\mathcal{K}^m$. This form comes with a corresponding notion of orthogonality. In particular, the well-known Gram-Schmidt orthogonalization procedure for a family of linearly independent vectors $\mathbf{b}_0, \dots, \mathbf{b}_{k-1} \in \mathcal{K}^m$ is defined as

$$\tilde{\mathbf{b}}_0 := \mathbf{b}_0, \quad \tilde{\mathbf{b}}_i := \mathbf{b}_i - \sum_{j=0}^{i-1} \frac{\langle \mathbf{b}_i, \tilde{\mathbf{b}}_j \rangle}{\langle \tilde{\mathbf{b}}_j, \tilde{\mathbf{b}}_j \rangle} \cdot \tilde{\mathbf{b}}_j.$$

One readily checks that $\langle \tilde{\mathbf{b}}_i, \tilde{\mathbf{b}}_j \rangle_\mathcal{K} = 0$. The generalized Gram-Schmidt norm of $\mathbf{B} \in \mathcal{K}^{m \times k}$ is defined as $\|\tilde{\mathbf{B}}\|_\mathcal{K} := \max_{i=0}^{k-1} \left\| \varphi \left(\langle \tilde{\mathbf{b}}_i, \tilde{\mathbf{b}}_i \rangle_\mathcal{K} \right) \right\|_\infty^{1/2}$.

NTRU lattices are a special class of lattices widely used in cryptography. Let $q \in \mathbb{Z}^+$.

Definition 5 (NTRU lattice). *Let $f, g, F, G \in \mathcal{R}$ satisfy NTRU equation*

$$fG - gF = q \pmod{(x^n + 1)}.$$

Then the NTRU lattice generated by f, g, F, G is the lattice generated by the columns of the block matrix

$$\mathbf{B}_{f,g,F,G} = \begin{pmatrix} \mathcal{A}(g) & \mathcal{A}(G) \\ \mathcal{A}(-f) & \mathcal{A}(-F) \end{pmatrix} \in \mathbb{Z}^{2n \times 2n}$$

where $\mathcal{A}(f) \in \mathbb{Z}^{n \times n}$ is the anti-cyclic matrix of polynomial $f \in \mathcal{R} = \mathbb{Z}[x]/(x^n + 1)$.

Given f and g , although there are many different pairs F and G that satisfy the NTRU equation, they all generate the same NTRU lattice.

Lemma 7 ([HHGP⁺03, Theorem 1]). *Given f, g , suppose that F, G and F', G' both satisfy NTRU equation. Then there is an element $c \in \mathcal{R}$ so that $F' = F + c \cdot f$ and $G' = G + c \cdot g$. Meanwhile, the Gram-Schmidt orthogonal basis of $\mathbf{B}_{f,g,F,G}$ and $\mathbf{B}_{f,g,F',G'}$ are same i.e.*

$$\tilde{F} = \tilde{F}' = \frac{qf^*}{ff^* + gg^*}, \tilde{G} = \tilde{G}' = \frac{qg^*}{ff^* + gg^*}.$$

3 GPV Signatures With Weak Smoothness

The signing process of a GPV signature scheme is typically implemented with lattice Gaussian sampling. The achieved Gaussian width, which determines both the signature size and the forgery security, is represented as

$$s = \mathbf{Q}(\mathbf{T}) \cdot \eta_\epsilon(\mathbb{Z}^m)$$

where $\mathbf{Q}(\mathbf{T})$ is the trapdoor quality depending on both the trapdoor itself and the sampler, and $\eta_\epsilon(\mathbb{Z}^m)$ is specified by the closeness parameter ϵ . Existing GPV signatures work with a *strong smoothness* condition, i.e., $\epsilon^{-1} = \omega(m^2)$.

This section investigates GPV signatures with *weak smoothness* where $\epsilon^{-1} = O(m)$ in the smoothing parameter setting, aiming for smaller size and higher security. We prove in Section 3.2 the strong unforgeability of GPV signatures with weak smoothness in the random oracle model under the original SIS and an additional assumption $\chi\text{-BDD}_{\mathcal{L}}^{\mathcal{O}_s}$ ($\chi\text{-BDD}_{\mathcal{L}}$ with Oracle) that is a variant of $\chi\text{-BDD}$ (Definition 2). Section 3.3 provides some cryptanalysis of $\chi\text{-BDD}_{\mathcal{L}}^{\mathcal{O}_s}$ and shows that this additional assumption has *no impact* on the concrete security of practical GPV signatures.

3.1 Closeness Parameter in Current GPV Signatures

Let us first recall the current GPV signature paradigm. In a GPV signature scheme, the public key is $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$, which is statistically near-uniform or computationally pseudorandom under certain assumptions, e.g., LWE or NTRU, while the secret key is a trapdoor $\mathbf{T}$ that provides a short basis of $\mathcal{L}^\perp(\mathbf{A})$. To sign a message $\mathbf{m}$, the signer first hashes $\mathbf{m}$ to a random syndrome $\mathbf{u} = \mathbf{H}(\mathbf{m})$ and

then uses $\mathbf{T}$ to compute a short preimage $\mathbf{s} \leftarrow D_{\mathcal{L}_{\mathbf{u}}^\perp(\mathbf{A}),s}$ by Gaussian sampling $\text{SampleG}(\mathcal{L}_{\mathbf{u}}^\perp(\mathbf{A}), s)$.

While smaller s implies better efficiency and security, it cannot be too small in order to ensure the provable security shown in [GPV08]. In more detail, the original security proof requires $s \geq \eta_\epsilon(\mathcal{L}^\perp(\mathbf{A}))$ for a negligible ϵ so that the following two distributions $\mathcal{P}_1$ and $\mathcal{P}_2$ are *statistically close* within distance 2ϵ by Lemma 2:

$$\begin{aligned}\mathcal{P}_1 &= \{(\mathbf{u}, \mathbf{s}) : \mathbf{u} \leftarrow U(\mathbb{Z}_q^n), \mathbf{s} \leftarrow D_{\mathcal{L}_{\mathbf{u}}^\perp(\mathbf{A}),s}\} \\ \mathcal{P}_2 &= \{(\mathbf{u}, \mathbf{s}) : \mathbf{s} \leftarrow D_{\mathbb{Z}^m,s}, \mathbf{u} = \mathbf{A}\mathbf{s} \bmod q\}.\end{aligned}$$

This allows us to simulate the signing process without knowing the trapdoor, which is crucial for the security proof. Additionally, existing Gaussian samplers need a sufficiently small ϵ to control the closeness between their sample distribution and the ideal Gaussian. In summary, the closeness parameter ϵ is related to the following two statistical distances:

- *Gaussian sampling closeness*:

$$\Delta(D_{\mathcal{L}_{\mathbf{u}}^\perp(\mathbf{A}),s}, \text{SampleG}(\mathcal{L}_{\mathbf{u}}^\perp(\mathbf{A}), s)) = O(\epsilon).$$

- *Uniform syndrome closeness* :

$$\Delta(U(\mathbb{Z}_q^n), \mathbf{u} = \mathbf{A} \cdot D_{\mathbb{Z}^m,s} \bmod q) = O(\epsilon).$$

In the past decade, a line of work [DLP14, BLRL⁺18, Pre17, MW17] uses various statistical tools and relaxes the condition on ϵ for *Gaussian sampling closeness*, which leads to a substantial improvement on practical GPV signatures. In particular, FALCON sets $\epsilon \leq 1/\sqrt{Q_s \cdot \lambda}$ following [Pre17] where $Q_s = 2^{64}$ denotes the number of signature queries and $\lambda = 128, 256$ is the target bit security. On the contrary, the ϵ requirement for *uniform distribution closeness* remains largely unexplored. As shown in [BLRL⁺18], using Rényi divergence allows to achieve the uniform syndrome closeness with some relaxed ϵ , but such ϵ should be of the same magnitude required by the Gaussian sampling closeness condition.

3.2 Security Proof for GPV Signatures with Weak Smoothness

We note that it is feasible to relax the condition on ϵ for *uniform syndrome closeness* as well, even if the distribution of simulated syndrome $\mathbf{u} = \mathbf{A} \cdot D_{\mathbb{Z}^m,s} \bmod q$ would be no longer *statistically close* to uniform. This section is dedicated to the security proof of GPV signatures with such relaxed uniform syndrome closeness.

Description of the scheme. Let Π denote the GPV signature scheme with weak smoothness, specified by the following algorithms:

- $\text{KeyGen}(1^\lambda)$: compute a (pseudo)random matrix $\bar{\mathbf{A}} \in \mathcal{R}_q^{n \times m}$ along with its trapdoor $\mathbf{T} \in \mathcal{R}^{m \times w}$. Return $(pk, sk) = (\bar{\mathbf{A}}, \mathbf{T})$.

- **Sign**($\mathbf{m}, sk$): First sample $\text{salt} \leftarrow \{0, 1\}^{2\lambda}$ and compute $\mathbf{u} = H(\mathbf{m}, \text{salt})$. Then sample $\mathbf{x} \leftarrow D_{\mathcal{L}_{\mathbf{u}}^\perp(\bar{\mathbf{A}}), s}$ using sk where $s \geq \eta_\epsilon(\mathcal{L}^\perp(\bar{\mathbf{A}}))$ with $\epsilon^{-1} = O(m)$. If $\|\mathbf{x}\| \leq \beta$, output $\text{sig} = (\text{salt}, \mathbf{x})$; otherwise restart.
- **Verfy**($\mathbf{m}, \text{sig}, pk$): Accept sig if $\|\mathbf{x}\| \leq \beta$ and $\bar{\mathbf{A}}\mathbf{x} = H(\mathbf{m}, \text{salt}) \bmod q$.

It suffices to consider the case where the public matrix $\bar{\mathbf{A}}$ in Π is in the HNF form, i.e. $\bar{\mathbf{A}} = [\mathbf{I}_n \mid \mathbf{A}]$.

New assumption: χ -BDD with Oracle. To give a security proof of Π , we introduce a new variant of the χ -BDD problem by adding an oracle outputting a Gaussian sample in some queried lattice coset.

Definition 6 (χ -BDD with Oracle $\chi\text{-BDD}_{\mathcal{L}}^{\mathcal{O}_s}(\{\mathbf{t}_i\}_i)$). Let $\chi : \mathbb{Z}^n \rightarrow [0, 1]$ be some distribution, $s > 0$ and $\mathcal{L} \subseteq \mathbb{Z}^n$ be a full-rank lattice. The χ -BDD with oracle problem $\chi\text{-BDD}_{\mathcal{L}}^{\mathcal{O}_s}(\{\mathbf{t}_i\}_i)$ is to determine whether $\{\mathbf{t}_i\}_i$ are sampled from $\chi \pmod{\mathcal{L}}$ or $U(\mathbb{Z}^n/\mathcal{L})$ with an oracle $\mathcal{O}_s$ that returns one vector sampled from $D_{\mathcal{L}+\mathbf{t}_i, s}$ for each $\mathbf{t}_i$. The χ -BDD with oracle assumption states that there exists no PPT adversary $\mathcal{A}$ that solves $\chi\text{-BDD}_{\mathcal{L}}^{\mathcal{O}_s}(\{\mathbf{t}_i\}_i)$ with non-negligible probability.

The rest of this section will focus on the $\chi\text{-BDD}_{\mathcal{L}}^{\mathcal{O}_s}(\{\mathbf{t}_i\}_i)$ instance with

$$\chi = D_{\mathcal{R}^m, s} \quad \text{and} \quad \mathcal{L} = \mathcal{L}^\perp(\bar{\mathbf{A}}).$$

The strong unforgeability of Π in the ROM is shown in Theorem 2.

Theorem 2. Let Π be a GPV signature scheme with weak smoothness and $\mathcal{S}$ be an adversary in the SUF-CMA security game against Π making at most Q_s signing queries and at most Q_H random queries. Let $\chi = D_{\mathcal{R}^m, s}$ and $\mathcal{L} = \mathcal{L}^\perp(\bar{\mathbf{A}})$. Denote an adversary $\mathcal{A}$'s advantage against $\chi\text{-BDD}_{\mathcal{L}}^{\mathcal{O}_s}(\{\mathbf{t}_i\}_i)$ by $\text{Adv}_{\mathcal{A}}^{\chi\text{-BDD}^{\mathcal{O}_s}}$ and an adversary $\mathcal{D}$'s advantage of distinguishing pk 's distribution from uniform by $\text{Adv}_{\mathcal{D}}^{pk}$. Then there exists an adversary $\mathcal{C}$ running in time $T_{\mathcal{C}} \approx T_{\mathcal{A}} \approx T_{\mathcal{D}} \approx T_{\mathcal{S}}$ against the SIS problem with advantage $\text{Adv}_{\mathcal{C}}^{\text{SIS}}$ such that

$$\text{Adv}_{\mathcal{S}}^{\text{SUF-CMA}} \leq \text{Adv}_{\mathcal{A}}^{\chi\text{-BDD}^{\mathcal{O}_s}} + \text{Adv}_{\mathcal{C}}^{\text{SIS}} + \text{Adv}_{\mathcal{D}}^{pk} + \text{negl}(\lambda).$$

Proof. We prove the security by intermediary hybrid games, starting from the SUF-CMA game against Π in the ROM and finally arriving at a game where we can build an adversary $\mathcal{C}$ against the SIS problem.

Game₀: This is the original SUF-CMA security game. A key pair $(pk, sk) \leftarrow \text{KeyGen}(1^\lambda)$ is generated. Given pk , $\mathcal{S}$ gets access to signing oracle which on the input $\mathbf{m}$ returns a signature sig and adds $(\mathbf{m}, \text{sig})$ to a table $\mathcal{T}_s$. Besides, the calls to the random oracle H are stored in a table $\mathcal{T}_H$.

Game₁: In this game, we answer oracle queries as follows.

- Random oracle($\mathbf{m}, \text{salt}$): If an entry has not been queried before, H returns $\mathbf{u} = (\bar{\mathbf{A}}\mathbf{x} \bmod q)$ where $\mathbf{x}$ is sampled from $D_{\mathcal{R}^m, s}$. We store in $\mathcal{T}_H$ the entry $((\mathbf{m}, \text{salt}), \mathbf{u})$.
- Signing oracle($\mathbf{m}$): Sample $\text{salt} \leftarrow \{0, 1\}^{2\lambda}$. Abort if an entry matching $(\mathbf{m}, \text{salt})$ exists in $\mathcal{T}_H$. Otherwise, return $\text{sig} = (\text{salt}, \mathbf{x})$ with $\mathbf{x}$ drawn from $D_{\mathcal{R}^m, s}$. Store $(\mathbf{m}, \text{sig})$ in $\mathcal{T}_s$ and $((\mathbf{m}, \text{salt}), \bar{\mathbf{A}}\mathbf{x} \bmod q)$ in $\mathcal{T}_H$.

If abort does not happen, then observe that under the $\chi\text{-BDD}_{\mathcal{L}^s}^{\mathcal{O}_s}(\{\mathbf{t}_i\}_i)$ assumption, the view of $\mathcal{S}$ in **Game₀** is indistinguishable from that in **Game₁**. Otherwise, a PPT algorithm $\mathcal{A}$ solving $\chi\text{-BDD}_{\mathcal{L}^s}^{\mathcal{O}_s}(\{\mathbf{t}_i\}_i)$ can be constructed as follows. Given a $\chi\text{-BDD}^{\mathcal{O}}$ instance with $\mathcal{L} = \mathcal{L}^\perp(\bar{\mathbf{A}})$ and the syndrome set $\{\mathbf{t}_i\}_i$, $\mathcal{A}$ uses $\bar{\mathbf{A}}$ as the public key and answers the random oracle by $\bar{\mathbf{A}}\mathbf{t}_i$ and the signing oracle by its oracle $\mathcal{O}_s(\mathbf{t}_i)$. Notice that algorithm $\mathcal{A}$ effectively interpolates between **Game₀** and **Game₁**. If the input to $\mathcal{A}$ is $\mathbf{t}_i \sim U(\mathcal{R}^m/\mathcal{L})$ for each i , then the view of $\mathcal{S}$ is just as in **Game₀**; otherwise the view of $\mathcal{S}$ is just as in **Game₁**. Thus

$$\left| \text{Adv}_{\mathcal{S}}^{\text{Game}_0} - \text{Adv}_{\mathcal{S}}^{\text{Game}_1} \right| \leq \text{Adv}_{\mathcal{A}}^{\chi\text{-BDD}^{\mathcal{O}_s}} + Q_s Q_H / 2^{2\lambda}.$$

Game₂: In this game, we replace the public matrix $\mathbf{A}$ by a uniformly random one. Clearly, by the pseudorandomness of $\mathbf{A}$, **Game₃** and **Game₄** are indistinguishable by $\mathcal{S}$, thus we have

$$\left| \text{Adv}_{\mathcal{S}}^{\text{Game}_1} - \text{Adv}_{\mathcal{S}}^{\text{Game}_2} \right| \leq \text{Adv}_{\mathcal{D}}^{pk}.$$

Game₃: Let $\text{sig}^* = (\text{salt}^*, \mathbf{x}^*) \notin \mathcal{T}_s$ be the forged signature output by $\mathcal{S}$ for a message $\mathbf{m}^*$. Without loss of generality, we assume that the pair $(\mathbf{m}^*, \text{salt}^*)$ has been queried to the random oracle H . From $\mathcal{T}_H$, retrieve $\hat{\mathbf{x}}$ corresponding to $(\mathbf{m}^*, \text{salt}^*)$. If $\mathbf{x}^* = \hat{\mathbf{x}}$, then abort. There are two cases to consider:

- If $H(\mathbf{m}^*, \text{salt}^*)$ was called by the signing oracle, then $\mathbf{x}^* \neq \hat{\mathbf{x}}$ since $\text{sig}^* = (\text{salt}^*, \mathbf{x}^*) \notin \mathcal{T}_s$. Then $\mathbf{x}^* - \hat{\mathbf{x}}$ is a solution to the SIS problem defined by the uniformly random $\mathbf{A}$.
- If $H(\mathbf{m}^*, \text{salt}^*)$ was queried directly to H , then by [JLWG24, Theorem 1.1] the conditional min-entropy of $\mathbf{x}^*$ given $\mathbf{u} = \bar{\mathbf{A}}\mathbf{x}^* \bmod q$ is at least $\omega(\log \lambda)$ and thus the probability of abort is negligible. Once no abort happens, the forgery implies a solution to the SIS problem.

So far, it follows that $\text{Adv}_{\mathcal{S}}^{\text{SUF-CMA}} \leq \text{Adv}_{\mathcal{A}}^{\chi\text{-BDD}^{\mathcal{O}_s}} + \text{Adv}_{\mathcal{C}}^{\text{SIS}} + \text{Adv}_{\mathcal{D}}^{pk} + \text{negl}(\lambda)$. $\square$

3.3 Cryptanalysis of $\chi\text{-BDD}^{\mathcal{O}_s}$

This section is dedicated to the justification for the hardness assumption $\chi\text{-BDD}^{\mathcal{O}_s}$ by concrete cryptanalysis. Our analysis focuses on the NTRU setting corresponding to our practical schemes in Section 5. Let $\mathcal{R} = \mathbb{Z}[x]/(x^n + 1)$, $(f, g) \in \mathcal{R}^2$ be the NTRU secret key and $h = g/f \bmod q$ be the NTRU public key. Let $\bar{\mathbf{A}} = [1 \mid h] \in \mathcal{R}_q^2$ and $\mathcal{L} = \mathcal{L}^\perp(\bar{\mathbf{A}})$ be the NTRU lattice. The length of (f, g) is about $\alpha\sqrt{q}$ for some $\alpha \approx 1.2$. Let $s = \alpha\sqrt{q} \cdot \eta_\varepsilon(\mathbb{Z})$ with $\varepsilon = \frac{1}{2n}$ and $\chi = D_{\mathcal{R}^2, s}$.

3.3.1 Lattice attack based analysis. A common method to solve χ -BDD is the lattice dual attack as discussed in [DP23]. Given the instance $\chi\text{-BDD}_{\mathcal{L}}^{\mathcal{O}_s}(\{\mathbf{t}_i\}_i)$, the oracle $\mathcal{O}_s$ is capable of outputting one extra short vector (v_0, v_1) from $D_{\mathcal{L}+\mathbf{t}_i, s}$, which is the distinction from the original $\chi\text{-BDD}_{\mathcal{L}}$. Note that the coefficients of $\mathbf{t}_i$ are of size $O(q)$ while the length of (v_0, v_1) is about $s\sqrt{2n} \approx \sqrt{n}\|(f, g)\|$. There seems no known technique to exploit these (v_0, v_1) 's to derive vectors shorter than trivial q -vectors in the dual lattice, thus the oracle $\mathcal{O}_s$ does not seem to be helpful to dual attacks. Next we ignore $\mathcal{O}_s$ and solely assess the hardness of χ -BDD. Figure 1 exhibits the cost of dual attacks solving χ -BDD: clearly, for our Gaussian parameter $s = 26.45$ (resp. 39.1) corresponding to $n = 512$ (resp. 1024), the hardness of χ -BDD is far higher than the underlying NTRU and SIS problems. Therefore, the additional assumption has *no impact* on the concrete security.

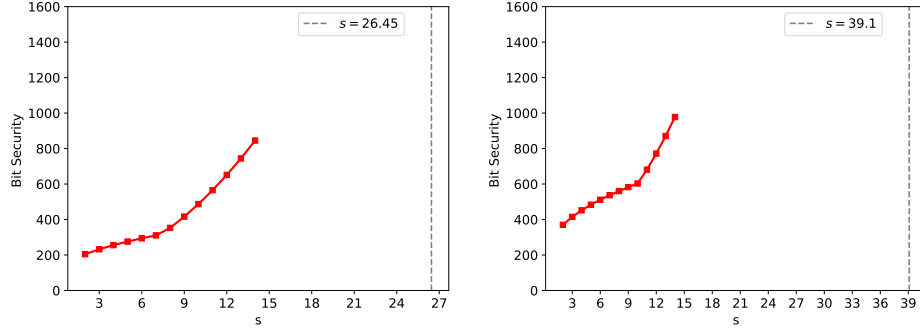


Fig. 1. The left graph is for $(n, q) = (512, 953)$ and the right one for $(1024, 1949)$.

3.3.2 Statistical analysis. In the context of NTRU, the distinguishing problem in χ -BDD is equivalent to distinguishing the distribution of $s_0 + hs_1 \bmod q$ with $(s_0, s_1) \leftarrow D_{\mathcal{R}^2, s}$ and the uniform distribution $U(\mathcal{R}_q)$. Suppose that the NTRU secret f is known, the problem can be further converted into distinguishing $fs_0 + gs_1 \bmod q$ with $(s_0, s_1) \leftarrow D_{\mathcal{R}^2, s}$ from uniform. We now provide some statistical evidence to demonstrate the intractability of this problem.

The following lemma gives a new bound for Rényi Divergence of Gaussian Sample over Λ/Λ' , which may have other potential applications.

Lemma 8. *Let Λ, Λ' be n -dimensional full-rank lattices with $\Lambda' \subseteq \Lambda$. Then for any $a \in (1, \infty)$, $\epsilon \in (0, 1)$, any $\sqrt{2}s \geq \eta_\epsilon(\Lambda')$ and $s \geq \sqrt{2} \cdot \eta_{\epsilon'}(\Lambda)$,*

$$R_a(U(\Lambda/\Lambda') \| D_{\Lambda, s, \mathbf{d}} \bmod \Lambda') \lesssim 1 + \frac{a(\epsilon + 4\epsilon')}{2}.$$

Proof. The first part proof is same with the relative error lemma in [Pre17]. Let $f_a : (x, y) \mapsto \frac{y^a}{(x+y)^{a-1}}$, then we use partial Taylor bounds for f_a on the point $(0, y)$. If $|x| \leq \delta \cdot y$, then

$$f_a(x, y) \leq f_a(0, y) + \frac{\partial f_a}{\partial x}(0, y) \cdot x + \frac{a(a-1)\delta^2}{2(1-\delta)^{a+1}} \cdot y.$$

Taking $y(\mathbf{c}) = \frac{\det(\Lambda)}{\det(\Lambda')}$, $x(\mathbf{c}) = \frac{\rho_s(\Lambda' + \mathbf{d} + \mathbf{c})}{\rho_s(\Lambda + \mathbf{d})} - \frac{\det(\Lambda)}{\det(\Lambda')}$, and $\frac{|x(\mathbf{c})|}{y(\mathbf{c})} = \delta_{\mathbf{c}} \leq \delta_{\max}$, then using the fact that $\sum_{\mathbf{c} \in \Lambda/\Lambda'} y(\mathbf{c}) = 1$ and $\sum_{\mathbf{c} \in \Lambda/\Lambda'} x(\mathbf{c}) = 0$ yields the result:

$$\begin{aligned} R_a(U(\Lambda/\Lambda') || D_{\Lambda, s} \bmod \Lambda')^{a-1} &= \sum_{\mathbf{c} \in \Lambda/\Lambda'} \frac{y(\mathbf{c})^a}{(x(\mathbf{c}) + y(\mathbf{c}))^{a-1}} \\ &\leq 1 + \sum_{\mathbf{c} \in \Lambda/\Lambda'} \frac{a(a-1)\delta_{\mathbf{c}}^2 \cdot y(\mathbf{c})}{2(1-\delta_{\mathbf{c}})^{a-1}} \\ &\leq 1 + \frac{a(a-1)}{2(1-\delta_{\max})^{a-1}} \cdot \sum_{\mathbf{c} \in \Lambda/\Lambda'} \delta_{\mathbf{c}}^2 \cdot y(\mathbf{c}). \end{aligned}$$

In the nest part proof, we analyze the upper bound of $\sum_{\mathbf{c} \in \Lambda/\Lambda'} \delta_{\mathbf{c}}^2 \cdot y(\mathbf{c})$.

$$\begin{aligned} \sum_{\mathbf{c} \in \Lambda/\Lambda'} \delta_{\mathbf{c}}^2 \cdot y(\mathbf{c}) &= \sum_{\mathbf{c} \in \Lambda/\Lambda'} \left(\frac{\det(\Lambda')}{\det(\Lambda)} \cdot \frac{\rho_s(\Lambda' + \mathbf{d} + \mathbf{c})}{\rho_s(\Lambda + \mathbf{d})} - 1 \right)^2 \cdot \frac{\det(\Lambda)}{\det(\Lambda')} \\ &= \left(\frac{\det(\Lambda')}{\det(\Lambda)} \right) \cdot \frac{\sum_{\mathbf{c} \in \Lambda/\Lambda'} \rho_s^2(\Lambda' + \mathbf{d} + \mathbf{c})}{\rho_s^2(\Lambda + \mathbf{d})} - 1. \end{aligned}$$

Using the equation in [RSD17]:

$$\rho_s(\Lambda + \mathbf{x})\rho_s(\Lambda + \mathbf{y}) = \sum_{\mathbf{c} \in \Lambda/(2\Lambda)} \rho_{\sqrt{2}s}(2\Lambda + \mathbf{c} + \mathbf{x} + \mathbf{y}) \cdot \rho_{\sqrt{2}s}(2\Lambda + \mathbf{c} + \mathbf{x} - \mathbf{y}),$$

we can get

$$\begin{aligned}
\sum_{\mathbf{c} \in \Lambda/\Lambda'} \rho_s^2(\Lambda' + \mathbf{d} + \mathbf{c}) &= \sum_{\mathbf{c} \in \Lambda/\Lambda'} \sum_{\mathbf{c}' \in \Lambda'/(2\Lambda')} \rho_{\sqrt{2}s}(2\Lambda' + \mathbf{c}' + 2\mathbf{d} + 2\mathbf{c}) \cdot \rho_{\sqrt{2}s}(2\Lambda' + \mathbf{c}') \\
&= \sum_{\mathbf{c}' \in \Lambda'/(2\Lambda')} \left(\sum_{\mathbf{c} \in \Lambda/\Lambda'} \rho_{\sqrt{2}s}(2\Lambda' + \mathbf{c}' + 2\mathbf{d} + 2\mathbf{c}) \right) \cdot \rho_{\sqrt{2}s}(2\Lambda' + \mathbf{c}') \\
&= \sum_{\mathbf{c}' \in \Lambda'/(2\Lambda')} \rho_{\sqrt{2}s}(2\Lambda + 2\mathbf{d} + \mathbf{c}') \cdot \rho_{\sqrt{2}s}(2\Lambda' + \mathbf{c}') \\
&\in [1 - \epsilon', 1 + \epsilon'] \cdot \frac{(\sqrt{2}s)^n}{\det(2\Lambda)} \cdot \sum_{\mathbf{c}' \in \Lambda'/(2\Lambda')} \rho_{\sqrt{2}s}(2\Lambda' + \mathbf{c}') \\
&\subseteq [1 - \epsilon', 1 + \epsilon'] \cdot \frac{(\sqrt{2}s)^n}{\det(2\Lambda)} \cdot \rho_{\sqrt{2}s}(\Lambda') \\
&\subseteq [(1 - \epsilon') \cdot (1 - \epsilon), (1 + \epsilon') \cdot (1 + \epsilon)] \cdot \frac{s^{2n}}{\det(\Lambda) \cdot \det(\Lambda')}.
\end{aligned}$$

Therefore, we have $\sum_{\mathbf{c} \in \Lambda/\Lambda'} \delta_{\mathbf{c}}^2 \cdot y(\mathbf{c}) \leq \epsilon + 4\epsilon'$. Finally, we can conclude that

$$R_a(U(\Lambda/\Lambda') || D_{\Lambda, s, \mathbf{d}} \bmod \Lambda') \leq \left(1 + \frac{a(a-1)(\epsilon + 4\epsilon')}{2(1 - \delta_{\max})^{a-1}} \right)^{\frac{1}{a-1}} \lesssim 1 + \frac{a(\epsilon + 4\epsilon')}{2}.$$

□

Lemma 9 shows that the inner product between a Gaussian vector and a given short vector follows some Gaussian.

Lemma 9 (Adapted from Theorem 3.3 [MP13]). *Let $\mathbf{z} \in \mathbb{Z}^m$ be a nonzero integer vector, $s \geq \sqrt{2}\|\mathbf{z}\|_\infty \cdot \eta_\epsilon(\mathbb{Z})$ and $\mathbf{c} = (c_1, \dots, c_m) \in \mathbb{R}^m$. Let $\mathbf{y} = (y_1, \dots, y_m)$ where y_i 's are independent samples from $D_{\mathbb{Z} + c_i, s}$ respectively. Then the distribution of $y = \langle \mathbf{z}, \mathbf{y} \rangle$ denoted as $D(y)$ is statistically close to $D_{Y, s'}$ where $Y = \gcd(\mathbf{z}) \cdot \mathbb{Z} + \langle \mathbf{z}, \mathbf{c} \rangle$ and $s' = s\|\mathbf{z}\|$. In addition, $R_\infty(D_{Y, s} || D(y)) \leq 1 + 2\epsilon$.*

The following is our main lemma. It shows that each individual coefficient of $fs_0 + gs_1 \bmod q$ is *statistically close* to uniform under certain condition which is satisfied by our schemes in Section 5. Under the heuristic assumption that all coefficients of $fs_0 + gs_1 \bmod q$ are independent distributed, one can even conclude that two distributions in $\chi\text{-BDD}^{O_s}$ are *statistically close*.

Lemma 10. *Given a vector $\mathbf{w} \in \mathbb{Z}^n$ and $\gcd(\mathbf{w}) = 1$, let $\mathbf{t} \leftarrow D_{\mathbb{Z}^n, s}$, then for $s \geq \sqrt{2}\|\mathbf{w}\|_\infty \cdot \eta_{\epsilon'}(\mathbb{Z})$ and $\sqrt{2}\|\mathbf{w}\|s \geq q \cdot \eta_\epsilon(\mathbb{Z})$, we have*

$$R_a(U(\mathbb{Z}_q) || \langle \mathbf{t}, \mathbf{w} \rangle \bmod q) \lesssim \left(1 + \frac{a(\epsilon + 4\epsilon')}{2} \right) \cdot (1 + 2\epsilon').$$

Remark 1. It is worth noting that ϵ' is associated with the smoothing parameter of $\mathbb{Z}$ and $\epsilon' < 2^{-100}$ even for the parameters of GPV signatures with weak smoothness.

Proof. Notice that $s \geq \sqrt{2}\|\mathbf{w}_q\|_\infty \cdot \eta_{\epsilon'}(\mathbb{Z})$ and $\gcd(\mathbf{w}_q) = 1$, therefore by Lemma 9, we have

$$R_\infty(D_{\mathbb{Z}, \|\mathbf{w}\|_s} \|D(\langle \mathbf{w}, \mathbf{t} \rangle)\|) \leq 1 + 2\epsilon'$$

for $\mathbf{t} \leftarrow D_{\mathbb{Z}^n, s}$. Also note that $s \geq \sqrt{2}\|\mathbf{w}\|_\infty \cdot \eta_{\epsilon'}(\mathbb{Z})$ and $\sqrt{2}\|\mathbf{w}\|_s \geq q \cdot \eta_\epsilon(\mathbb{Z})$, therefore by Lemma 8, we have

$$R_a(U(\mathbb{Z}_q) \| D_{\mathbb{Z}, \|\mathbf{w}_q\|_s} \bmod q) \lesssim 1 + \frac{a(\epsilon + 4\epsilon')}{2}.$$

Finally, we can get

$$\begin{aligned} & R_a(U(\mathbb{Z}_q) \| \langle \mathbf{t}, \mathbf{w} \rangle \bmod q) \\ & \leq R_a(U(\mathbb{Z}_q) \| D_{\mathbb{Z}, \|\mathbf{w}\|_s} \bmod q) \cdot R_\infty(D_{\mathbb{Z}, \|\mathbf{w}\|_s} \| D(\langle \mathbf{w}, \mathbf{t} \rangle)\|) \\ & \lesssim \left(1 + \frac{a(\epsilon + 4\epsilon')}{2}\right) \cdot (1 + 2\epsilon'). \end{aligned}$$

□

4 Preimage Sampler with Weak Smoothness

Section 3 has provided a theoretical security proof for the GPV signatures with weak smoothness, in which we assume the preimage sampler can sample from a Gaussian with *negligible* Gaussian sampling closeness. However, regular Klein-GPV sampler [GPV08] and Peikert sampler [Pei10] cannot fulfil this requirement and the use of a large closeness parameter possibly leaks the trapdoor information.

In this section, we present some modified Gaussian samplers compatible with the weak smoothness setting. Our idea is to apply rejection sampling techniques to rectify the discrepancy between the output distribution and the ideal Gaussian. Section 4.1 shows a modified Klein-GPV sampler which was originally suggested in [BLP⁺13] for security reduction. The rejection sampling for Klein-GPV sampler is not directly applicable to Peikert sampler. Then, we propose two Peikert samplers with weak smoothness in Section 4.2 with some different treatments. Section 4.3 further presents a modified hybrid sampler [Pre15] under weak smoothness condition and we give an accurate estimate of the expected number of repetitions. Furthermore, Section 4.4 describes an efficient algorithm to compute the division of theta functions over integer lattices which is a crucial step in the rejection sampling.

4.1 Klein-GPV Sampler with Weak Smoothness

The formal description of the Klein-GPV sampler with weak smoothness is given by Algorithm 1 that was first proposed in [BLP⁺13].

It should be emphasized that the output of Algorithm 1 *exactly* follows the target distribution. The expected number of repetitions is closely related to the closeness parameter ϵ , which is bounded by some constant when $\epsilon = O(\frac{1}{n})$.

Algorithm 1: The Modified KGPV sampler

Input: A basis $\mathbf{B} = (\mathbf{b}_0, \dots, \mathbf{b}_{n-1})$, a center $\mathbf{c}$ and $s \geq \|\mathbf{B}\|_{GS} \cdot \eta_\epsilon(\mathbb{Z})$

Output: A lattice point $\mathbf{v}$ exactly following $D_{\mathcal{L}(\mathbf{B}), s, \mathbf{c}}$

```

1  $\mathbf{v} \leftarrow \mathbf{0}, \mathbf{c}' \leftarrow \mathbf{c}, \Delta \leftarrow 1$ 
2 for  $i = n-1, \dots, 0$  do
3    $\mathbf{c}_i'' \leftarrow \langle \mathbf{c}', \tilde{\mathbf{b}}_i \rangle / \|\tilde{\mathbf{b}}_i\|^2, s_i \leftarrow s / \|\tilde{\mathbf{b}}_i\|$ 
4    $z_i \leftarrow D_{\mathbb{Z}, s_i, \mathbf{c}_i''}$ 
5    $\Delta \leftarrow \Delta \cdot \frac{\rho_{s_i, \mathbf{c}_i''}(\mathbb{Z})}{\rho_{s_i}(\mathbb{Z})}$ 
6    $\mathbf{c}' \leftarrow \mathbf{c}' - z_i \mathbf{b}_i, \mathbf{v} \leftarrow \mathbf{v} + z_i \mathbf{b}_i$ 
7 end for
8 return  $\mathbf{v}$  with probability  $\Delta$  otherwise restart

```

Lemma 11. *Algorithm 1 is correct and the expected number of repetitions is at most $\left(\frac{1+\epsilon}{1-\epsilon}\right)^n$.*

Proof. Lemma 1 indicates that $\Delta \leq 1$ and

$$\Delta = \prod_{i=0}^{n-1} \frac{\rho_{s_i, \mathbf{c}_i''}(\mathbb{Z})}{\rho_{s_i}(\mathbb{Z})} \geq \left(\frac{1-\epsilon}{1+\epsilon}\right)^n.$$

Thus the expected number of repetitions is bounded by $\left(\frac{1+\epsilon}{1-\epsilon}\right)^n$.

Algorithm 1 is the same with regular Klein-GPV sampler except for Steps 5 and 8 for the rejection sampling. As shown in [GPV08, Lemma 4.5], the probability that Klein-GPV sampler outputs $\mathbf{z}$ is exactly

$$P'(\mathbf{z}) = \frac{\rho_{s, \mathbf{c}}(\mathbf{z})}{\prod_{i=0}^{n-1} \rho_{s_i, \mathbf{c}_i''}(\mathbb{Z})}.$$

After rejection sampling is added, the probability that algorithm 1 outputs $\mathbf{z}$ is

$$P(\mathbf{z}) = P'(\mathbf{z}) \cdot \frac{\prod_{i=0}^{n-1} \rho_{s_i, \mathbf{c}_i''}(\mathbb{Z})}{\prod_{i=0}^{n-1} \rho_{s_i}(\mathbb{Z})} = \frac{\rho_{s, \mathbf{c}}(\mathbf{z})}{\prod_{i=0}^{n-1} \rho_{s_i}(\mathbb{Z})} \propto \rho_{s, \mathbf{c}}(\mathbf{z}).$$

Then the proof is completed. $\square$

Remark 2. The repetition number in Algorithm 1 may leak $\prod_{i=0}^{n-1} \rho_{s_i}(\mathbb{Z})$ where s_i 's are secret-dependent. While the knowledge of all s_i 's may give rise to key recovery attacks [FKT⁺20], leveraging $\prod_{i=0}^{n-1} \rho_{s_i}(\mathbb{Z})$ solely to mount an effective attack seems highly challenging at present. We can completely seal this leak by the same technique of [HPRR20]: selecting a universal $P \geq \prod_{i=0}^{n-1} \rho_{s_i}(\mathbb{Z})$ and tweaking Δ to $\Delta \cdot \frac{\prod_{i=0}^{n-1} \rho_{s_i}(\mathbb{Z})}{P}$.

Algorithm 2: The Modified Peikert sampler

Input: A basis $\mathbf{B} \in \mathbb{Z}^{n \times n}$, a rounding parameter $r \geq \eta_\epsilon(\mathbb{Z}^n)$, a center $\mathbf{c}$, a matrix $\Sigma > \Sigma_1 = r^2 \mathbf{B} \mathbf{B}^t$ and a matrix $\mathbf{C} = \sqrt{\Sigma_2}$ where $\Sigma_2 = \Sigma - \Sigma_1 > 0$.

Output: A lattice point $\mathbf{z}$ exactly following $D_{\mathcal{L}(\mathbf{B}), \sqrt{\Sigma}, \mathbf{c}}$.

- 1 $\mathbf{x} \leftarrow \mathcal{N}_{\sqrt{\Sigma_2}}$
- 2 $\mathbf{c}' \leftarrow \mathbf{B}^{-1}(\mathbf{c} - \mathbf{x})$
- 3 accept $\mathbf{x}$ with probability $\frac{\rho_{r, \mathbf{c}'}(\mathbb{Z}^n)}{\rho_r(\mathbb{Z}^n)}$ otherwise restart
- 4 **return** $\mathbf{z} \leftarrow \mathbf{B} \lfloor \mathbf{c}' \rfloor_r$

4.2 Peikert Sampler with Weak Smoothness

In this section, we adapt Peikert sampler to the weak smoothness setting. Our goal is to achieve a Gaussian width $\eta_\epsilon(\mathbb{Z}^n) \cdot s_1(\mathbf{B})$ with a large ϵ , say $\epsilon = O(1)$. The adaption is not so straightforward as the case of Klein-GPV sampler, as Peikert sampler contains two phases: perturbation sampling and online sampling, and some extra condition needs to be taken into account.

We start with a relatively simpler case where the perturbation vector is drawn from a continuous Gaussian. Algorithm 2 gives a formal description. Its correctness is given in Lemma 12.

Lemma 12. *Algorithm 2 is correct and the expected number of repetitions is at most $\frac{1+\epsilon}{1-\epsilon}$.*

Proof. By Lemma 1, $\frac{\rho_{r, \mathbf{c}'}(\mathbb{Z}^n)}{\rho_r(\mathbb{Z}^n)} \geq \frac{1-\epsilon}{1+\epsilon}$, thus the expected number of repetitions is bounded by $\frac{1+\epsilon}{1-\epsilon}$. Let **Out** denote the output of Algorithm 2, then

$$\Pr[\text{Out} = \mathbf{z}] = \int_{\mathbb{R}^n} \frac{\rho_{\sqrt{\Sigma_2}}(\mathbf{x})}{\sqrt{\det(\Sigma_2)}} \cdot \frac{\rho_{r, \mathbf{c}'}(\mathbb{Z}^n)}{\rho_r(\mathbb{Z}^n)} \cdot \frac{\rho_{\sqrt{\Sigma_1}}(\mathbf{z} - (\mathbf{c} - \mathbf{x}))}{\rho_{\sqrt{\Sigma_1}, \mathbf{c} - \mathbf{x}}(\mathcal{L}(\mathbf{B}))} d\mathbf{x}.$$

By Lemma 3, we have

$$\rho_{\sqrt{\Sigma_2}}(\mathbf{x}) \rho_{\sqrt{\Sigma_1}}(\mathbf{z} - (\mathbf{c} - \mathbf{x})) = \rho_{\sqrt{\Sigma}}(\mathbf{z} - \mathbf{c}) \rho_{\sqrt{\Sigma'}}(\mathbf{x} - \mathbf{c}'')$$

for some $\mathbf{c}''$ and Σ' . For $\mathbf{y} \leftarrow D_{\mathbb{Z}^n, r, \mathbf{B}^{-1}(\mathbf{c} - \mathbf{x})}$, the distribution of $\mathbf{z} = \mathbf{B}\mathbf{y}$ is $D_{\mathcal{L}(\mathbf{B}), r\mathbf{B}, \mathbf{c} - \mathbf{x}}$. Then

$$\begin{aligned} \Pr[\text{Out} = \mathbf{z}] &= \frac{\rho_{\sqrt{\Sigma}}(\mathbf{z} - \mathbf{c})}{\sqrt{\det(\Sigma_2)} \cdot \rho_r(\mathbb{Z}^n)} \int_{\mathbb{R}^n} \rho_{\sqrt{\Sigma'}}(\mathbf{x} - \mathbf{c}'') d\mathbf{x} \\ &= \frac{\rho_{\sqrt{\Sigma}}(\mathbf{z} - \mathbf{c})}{\sqrt{\det(\Sigma_2)} \cdot \rho_r(\mathbb{Z}^n)} \cdot \sqrt{\det(\Sigma')} \\ &\propto \rho_{\sqrt{\Sigma}}(\mathbf{z} - \mathbf{c}). \end{aligned}$$

The correctness follows and we complete the proof. $\square$

Algorithm 3: The Modified Peikert sampler over q -ary lattice

Input: A basis $\mathbf{B} \in \mathbb{Z}^{n \times n}$ of q -ary lattice, a rounding parameter $r \geq \eta_\epsilon(\mathbb{Z}^n)$, a center $\mathbf{c}$, a matrix $\Sigma \geq r^2(\mathbf{B}\mathbf{B}^t) + \mathbf{I}$, $\Sigma_1 = r^2\mathbf{B}\mathbf{B}^t$, a matrix $\Sigma_2 = \Sigma - \Sigma_1 > \mathbf{I}$, a matrix $\mathbf{Z} = q \cdot \mathbf{B}^{-1} \in \mathbb{Z}^{n \times n}$, a scaling parameter $L \in \mathbb{Z}$ satisfies $L\sqrt{\Sigma_3} \geq \eta_{\epsilon'}(\mathbb{Z}^n)$ where matrix $\Sigma_3^{-1} = \Sigma_1^{-1} + \Sigma_2^{-1}$.

Output: A lattice point $\mathbf{z}$ following a distribution close to $D_{\mathcal{L}(\mathbf{B}), \sqrt{\Sigma}, \mathbf{c}}$.

```

1  $\mathbf{x}' \leftarrow D_{\mathbb{Z}^n, L\sqrt{\Sigma_2}}$ 
2  $\mathbf{x} \leftarrow \frac{1}{L} \cdot \mathbf{x}'$ 
3  $\mathbf{c}' \leftarrow \frac{1}{q} \cdot \mathbf{Z}(\mathbf{c} - \mathbf{x})$ 
4 accept  $\mathbf{x}$  with probability  $\frac{\rho_{r, \mathbf{c}'}(\mathbb{Z}^n)}{\rho_r(\mathbb{Z}^n)}$  otherwise restart
5 return  $\mathbf{z} \leftarrow \mathbf{B}[\mathbf{c}']_r$ 

```

Next we consider the case of discrete perturbation, in which the Σ_3 -condition requires extra care to guarantee the accuracy of output. To this end, we introduce two additional parameters ϵ' and L , where ϵ' controls the closeness between the output and ideal distribution and the integer L is the scaling factor to densify the perturbation lattice to satisfy the Σ_3 -condition. The detailed algorithmic description is given in Algorithm 3.

Remark 3. The sampling of $\mathbf{x}' \leftarrow D_{\mathbb{Z}^n, L\sqrt{\Sigma_2}}$ can be implemented with Peikert's convolution technique using continuous Gaussian [Pei10] or the floating-point free perturbation sampling technique [DGPY20]. We omit its implementation details and the discrepancy between the sampled distribution and the ideal one.

Lemma 13. *Algorithm 3 is correct. If $\epsilon' \leq 2^{-8}$, the distribution of output $\mathbf{z}$ is within statistical distance $6\epsilon'$ with $D_{\mathcal{L}(\mathbf{B}), \sqrt{\Sigma}, \mathbf{c}}$ and the expected number of repetitions is at most $\frac{1+\epsilon}{1-\epsilon}$.*

Proof. Let Out denote the output of Algorithm 3 and X be the random variable output at Step 2, then

$$\Pr[\text{Out} = \mathbf{z} \wedge X = \mathbf{x}] = \frac{\rho_{\sqrt{\Sigma_2}}(\mathbf{x})}{\rho_{\sqrt{\Sigma_2}}(\frac{1}{L} \cdot \mathbb{Z}^n)} \cdot \frac{\rho_{r, \mathbf{c}'}(\mathbb{Z}^n)}{\rho_r(\mathbb{Z}^n)} \cdot \frac{\rho_{\sqrt{\Sigma_1}}(\mathbf{z} - (\mathbf{c} - \mathbf{x}))}{\rho_{\sqrt{\Sigma_1}, \mathbf{c} - \mathbf{x}}(\mathcal{L}(\mathbf{B}))}.$$

By the same argument in Lemma 12, it follows that

$$\begin{aligned} \Pr[\text{Out} = \mathbf{z}] &= \frac{\rho_{\sqrt{\Sigma}}(\mathbf{z} - \mathbf{c})}{\rho_{\sqrt{\Sigma_2}}(\frac{1}{L} \cdot \mathbb{Z}^n) \cdot \rho_r(\mathbb{Z}^n)} \cdot \rho_{\sqrt{\Sigma_3}}\left(\frac{1}{L} \cdot \mathbb{Z}^n - \mathbf{c}''\right) \\ &\in \left[\frac{1-\epsilon'}{1+\epsilon'}, \frac{1+\epsilon'}{1-\epsilon'}\right] \cdot \frac{\rho_{\sqrt{\Sigma}}(\mathbf{z} - \mathbf{c})}{\rho_{\sqrt{\Sigma_2}}(\frac{1}{L} \cdot \mathbb{Z}^n) \cdot \rho_r(\mathbb{Z}^n)} \cdot \rho_{\sqrt{\Sigma_3}}\left(\frac{1}{L} \cdot \mathbb{Z}^n\right) \\ &\propto \left[\frac{1-\epsilon'}{1+\epsilon'}, \frac{1+\epsilon'}{1-\epsilon'}\right] \cdot \rho_{\sqrt{\Sigma}}(\mathbf{z} - \mathbf{c}). \end{aligned}$$

The second line is by $\sqrt{\Sigma_3} \geq \frac{1}{L} \cdot \eta_{\epsilon'}(\mathbb{Z}^n)$ and Lemma 1. A routine computation yields that

$$\Pr[\text{Out} = \mathbf{z}] \in [1 - 6\epsilon', 1 + 6\epsilon'] \cdot D_{\mathcal{L}(\mathbf{B}), \sqrt{\Sigma}, \mathbf{c}}(\mathbf{z}).$$

Again, by the same argument in Lemma 12, the upper bound of the expected number of repetitions follows. So far, the proof is completed. $\square$

Remark 4. The rejection sampling in Algorithm 2 and Algorithm 3 may leak $\det(\Sigma_2)$ through the acceptance rate. Similar to the case in Remark 2, this leakage is difficult to exploit and can be easily mitigated.

4.3 Hybrid Sampler with Weak Smoothness

In the conference version of this paper [ZJR⁺25], the adaption of MITAKA and ANTRAG to weak smoothness was left as an open question. Both MITAKA and ANTRAG use the Prest's hybrid sampler [Pre15] to reconcile security, efficiency and simplicity. The hybrid sampler can be seen as the randomization of Babai's nearest plane algorithm over rings, which is similar to the Klein-GPV sampler, but the randomization in the hybrid sampler is performed at the ring level, which is implemented by Peikert sampler in a parallel manner.

In this section, we resolve this open question by presenting a modified hybrid sampler that is compatible with the weak smoothness condition. Our main observation is that following the approach of Algorithm 2, applying rejection sampling to the rounding $\lfloor \mathbf{c}' \rfloor_r$ ensures that the normalization constant of Peikert sampling over rings becomes independent of the output vector, which allows us to bypass the intractable calculation of the high-dimensional ellipsoid Gaussian mass. Interestingly, one rejection sampling is sufficient to the modified hybrid sampler. The detailed algorithm is shown in Algorithm 4. In order to maintain consistency with the matrix-based description used in aforementioned algorithms, all elements in $\mathcal{K} = \mathbb{Q}[x]/(x^n + 1)$ are represented in the matrix form.

Lemma 14. *Algorithm 4 is correct and the expected number of repetitions is at most $\left(\frac{1+\epsilon}{1-\epsilon}\right)^{nk}$.*

Proof. Lemma 1 indicates that $\Delta \leq 1$ and

$$\Delta = \prod_{i=0}^{n-1} \frac{\rho_{r, \mathbf{y}_i}(\mathbb{Z}^n)}{\rho_r(\mathbb{Z}^n)} \geq \left(\frac{1-\epsilon}{1+\epsilon}\right)^{nk}.$$

Thus the expected number of repetitions is bounded by $\left(\frac{1+\epsilon}{1-\epsilon}\right)^{nk}$.

Let Out denote the output of Algorithm 4 that $\mathbf{v} = \sum_{i=0}^{k-1} \mathbf{B}_i \mathbf{z}_i$. For each iteration i , let $P(\mathbf{z}_i, \mathbf{x}_i)$ denote the joint probability of $(\mathbf{z}_i, \mathbf{x}_i)$ induced by the “Peikert steps” (Steps 5-7) in that iteration. By Lemma 3, we obtain

$$\begin{aligned} P(\mathbf{z}_i, \mathbf{x}_i) &= \frac{\rho_{\sqrt{\Sigma_i - r^2 \mathbf{I}_n}}(\mathbf{x}_i)}{\det(\sqrt{\Sigma_i - r^2 \mathbf{I}_n})} \cdot \frac{\rho_r(\mathbf{z}_i - (\mathbf{d}_i - \mathbf{x}_i))}{\rho_{r, \mathbf{d}_i - \mathbf{x}_i}(\mathbb{Z}^n)} \\ &= \frac{\rho_{\sqrt{\Sigma_i}}(\mathbf{z}_i - \mathbf{d}_i)}{\det(\sqrt{\Sigma_i - r^2 \mathbf{I}_n})} \cdot \frac{\rho_{\sqrt{\Sigma_i'}}(\mathbf{x}_i - \mathbf{c}_i')}{\rho_{r, \mathbf{d}_i - \mathbf{x}_i}(\mathbb{Z}^n)} \end{aligned}$$

Algorithm 4: The Modified Hybrid sampler

Input: Basis $\mathbf{B} = \{\mathbf{B}_0, \dots, \mathbf{B}_{n-1}\} \in \mathbb{Z}^{nd \times nk}$ and its block GSO $\tilde{\mathbf{B}} = (\tilde{\mathbf{B}}_0, \dots, \tilde{\mathbf{B}}_{n-1}) \in \mathbb{R}^{nd \times nk}$, a center $\mathbf{c} \in \mathbb{Z}^{nd}$, a rounding parameter $r \geq \eta_\epsilon(\mathbb{Z})$ and $s > r \cdot \|\tilde{\mathbf{B}}\|_{\mathcal{K}}$
Output: a lattice point $\mathbf{v}$ exactly following $D_{\mathcal{L}(\mathbf{B}), s, \mathbf{c}}$.

```

1  $\mathbf{v}_{k-1} \leftarrow \mathbf{0}, \mathbf{c}_{k-1} \leftarrow \mathbf{c}, \Delta \leftarrow 1$ 
2 for  $i = k-1, \dots, 0$  do
3    $\mathbf{d}_i \leftarrow \tilde{\mathbf{B}}_i^+ \mathbf{c}_i$ 
4    $\Sigma_i \leftarrow s^2 (\tilde{\mathbf{B}}_i^t \tilde{\mathbf{B}}_i)^{-1}$ 
5    $\mathbf{x}_i \leftarrow \mathcal{N}_{\sqrt{\Sigma_i - r^2 \mathbf{I}_n}}$ 
6    $\mathbf{y}_i \leftarrow \mathbf{d}_i - \mathbf{x}_i$ 
7    $\mathbf{z}_i \leftarrow \lfloor \mathbf{y}_i \rfloor_r$ 
8    $\mathbf{c}_{i-1} \leftarrow \mathbf{c}_i - \mathbf{B}_i \mathbf{z}_i$ 
9    $\mathbf{v}_{i-1} \leftarrow \mathbf{v}_i + \mathbf{B}_i \mathbf{z}_i$ 
10   $\Delta \leftarrow \Delta \cdot \frac{\rho_{r, \mathbf{y}_i}(\mathbb{Z}^n)}{\rho_r(\mathbb{Z}^n)}$ 
11 end for
12 return  $\mathbf{v} = \mathbf{v}_0$  with probability  $\Delta$  otherwise restart

```

where the explicit forms of $\mathbf{c}'_i$ and Σ'_i are immaterial for our purposes. By the choice of Σ_i and the Gram-Schmidt decomposition (over $\mathcal{K}$) $\mathbf{B} = \tilde{\mathbf{B}}\mathbf{U}$, it follows that

$$\rho_s(\mathbf{v} - \mathbf{c}) = \prod_{i=0}^{k-1} \rho_{\sqrt{\Sigma_i}}(\mathbf{z}_i - \mathbf{d}_i).$$

We now evaluate

$$\begin{aligned}
\Pr[\text{Out} = \mathbf{v}] &= \int_{\mathbb{R}^n} \cdots \int_{\mathbb{R}^n} \prod_{i=0}^{k-1} \Delta \cdot P(\mathbf{z}_i, \mathbf{x}_i) d\mathbf{x}_0 \cdots d\mathbf{x}_{k-1} \\
&= \int_{\mathbb{R}^n} \cdots \int_{\mathbb{R}^n} \prod_{i=0}^{k-1} \frac{\rho_{r, \mathbf{y}_i}(\mathbb{Z}^n)}{\rho_r(\mathbb{Z}^n)} \cdot \frac{\rho_{\sqrt{\Sigma_i}}(\mathbf{z}_i - \mathbf{d}_i)}{\det(\sqrt{\Sigma_i - r^2 \mathbf{I}_n})} \cdot \frac{\rho_{\sqrt{\Sigma'_i}}(\mathbf{x}_i - \mathbf{c}'_i)}{\rho_{r, \mathbf{d}_i - \mathbf{x}_i}(\mathbb{Z}^n)} d\mathbf{x}_0 \cdots d\mathbf{x}_{k-1} \\
&= \int_{\mathbb{R}^n} \cdots \int_{\mathbb{R}^n} \prod_{i=0}^{k-1} \frac{\rho_{\sqrt{\Sigma_i}}(\mathbf{z}_i - \mathbf{d}_i)}{\det(\sqrt{\Sigma_i - r^2 \mathbf{I}_n}) \cdot \rho_r(\mathbb{Z}^n)} \cdot \rho_{\sqrt{\Sigma'_i}}(\mathbf{x}_i - \mathbf{c}'_i) d\mathbf{x}_0 \cdots d\mathbf{x}_{k-1} \\
&= \frac{\rho_{\sqrt{\Sigma}}(\mathbf{v} - \mathbf{c})}{\prod_{i=0}^{k-1} \det(\sqrt{\Sigma_i - r^2 \mathbf{I}_n}) \cdot \rho_r(\mathbb{Z}^n)} \cdot \prod_{i=0}^{k-1} \int_{\mathbb{R}^n} \rho_{\sqrt{\Sigma'_i}}(\mathbf{x}_i - \mathbf{c}'_i) d\mathbf{x}_i \\
&= \rho_s(\mathbf{v} - \mathbf{c}) \cdot \prod_{i=0}^{k-1} \frac{\det(\sqrt{\Sigma'_i})}{\det(\sqrt{\Sigma_i - r^2 \mathbf{I}_n}) \cdot \rho_r(\mathbb{Z}^n)} \\
&\propto \rho_s(\mathbf{v} - \mathbf{c}).
\end{aligned}$$

The correctness follows and we complete the proof. $\square$

Remark 5. The hybrid samplers in MITAKA and ANTRAG are instantiated over NTRU lattices with $d = k = 2$. In fact, the same hybrid sampling technique can also be applied to module-NTRU lattices [CPS⁺20].

4.3.1 Tight Estimate of Rejection Rate. The rejection sampling yields some loss in efficiency. Lemma 14 gives an upper bound of the expected repetition number, but that bound is overestimated. For notational convenience, we collect the fractional parts of all vectors $\mathbf{y}_i$ appearing in Algorithm 4 into a single vector $\mathbf{w} \in \mathbb{R}^{nk}$, i.e., $\mathbf{w} = (\mathbf{y}_0 - \lfloor \mathbf{y}_0 \rfloor, \dots, \mathbf{y}_{k-1} - \lfloor \mathbf{y}_{k-1} \rfloor)$. Because $r = \eta_\epsilon(\mathbb{Z})$ in the sampling procedure, the intermediate centers w_i tend to be uniform over $(-1/2, 1/2]$ and do not always fall into the worst-case regime—in particular, they do not necessarily make $\rho_{r, w_i}(\mathbb{Z})$ approach $1 - \epsilon$. To obtain a more refined estimate of the rejection probability for the variant of ANTRAG (in Section 5.3), we use the following assumption.

– *Heuristic:* $w_i \sim U((-1/2, 1/2])$ for $0 \leq i < nk$.

There is no trivial way to describe the probability distribution of the w_i 's. As a rough approximation, one may regard it as the reduction modulo 1 of a convolution between an approximately uniform distribution and a Gaussian distribution. Lemma 15 shows our tight estimate of the repetition number. The experimental results show that our heuristic captures the empirical behavior sufficiently well for the purpose of estimating the rejection rate.

Lemma 15. *Under the above heuristic, the expected repetition number M of Algorithm 4 for $\epsilon \in (0, 1)$ is upper-bounded by*

$$\left(1 + \frac{\epsilon}{1 + \epsilon} + \frac{4\epsilon^4(1 + \epsilon)}{2^5(1 + \epsilon)^5 - \epsilon^5}\right)^{nk}.$$

Proof. By Lemma 14 and the definition of $\mathbf{w}$, we have

$$M = \prod_{i=0}^{k-1} \frac{\rho_r(\mathbb{Z}^n)}{\rho_{r, \mathbf{y}_i}(\mathbb{Z}^n)} = \prod_{i=0}^{nk-1} \frac{\rho_r(\mathbb{Z})}{\rho_{r, w_i}(\mathbb{Z})}.$$

Under the heuristic that w_i is uniformly distributed over $(-1/2, 1/2]$, we have

$$\begin{aligned} \mathbb{E}[\rho_{r, w_i}(\mathbb{Z})] &= \int_{-\frac{1}{2}}^{\frac{1}{2}} \sum_{x \in \mathbb{Z}} \exp\left(-\pi \frac{(x - w_i)^2}{r^2}\right) dw_i \\ &= \sum_{x \in \mathbb{Z}} \int_{-\frac{1}{2}}^{\frac{1}{2}} \exp\left(-\pi \frac{(x - w_i)^2}{r^2}\right) dw_i \\ &= \int_{-\infty}^{+\infty} \exp\left(-\pi \frac{u^2}{r^2}\right) du = r. \end{aligned}$$

Next, we derive an explicit upper bound on $\rho_r(\mathbb{Z})$. Let $r = \overline{\eta}_\epsilon(\mathbb{Z})$, then

$$\begin{aligned}
\rho_r(\mathbb{Z}) &= r \cdot \rho_{1/r}(\mathbb{Z}) \\
&= r \cdot \left(1 + 2 \sum_{l \in \mathbb{N}^+} \left(\frac{\epsilon}{2(1+\epsilon)} \right)^{l^2} \right) \\
&\leq r \cdot \left(1 + \frac{\epsilon}{1+\epsilon} + 2 \left(\frac{\epsilon}{2(1+\epsilon)} \right)^4 \cdot \sum_{l \in \mathbb{N}^+} \left(\frac{\epsilon}{2(1+\epsilon)} \right)^{5l} \right) \\
&= r \cdot \left(1 + \frac{\epsilon}{1+\epsilon} + 2 \left(\frac{\epsilon}{2(1+\epsilon)} \right)^4 \cdot \frac{1}{1 - \left(\frac{\epsilon}{2(1+\epsilon)} \right)^5} \right) \\
&= r \cdot \left(1 + \frac{\epsilon}{1+\epsilon} + \frac{4\epsilon^4(1+\epsilon)}{2^5(1+\epsilon)^5 - \epsilon^5} \right).
\end{aligned}$$

The inequality follows from the fact that $x^{(l+2)^2} \leq x^{4+5l}$ for $x \in (0, 1)$ and $l \geq 1$. Then

$$\mathbb{E}[M] = \prod_{i=0}^{nk-1} \frac{\rho_r(\mathbb{Z})}{\mathbb{E}[\rho_{r, w_i}(\mathbb{Z})]} \leq \left(1 + \frac{\epsilon}{1+\epsilon} + \frac{4\epsilon^4(1+\epsilon)}{2^5(1+\epsilon)^5 - \epsilon^5} \right)^{nk}$$

and the proof is completed. $\square$

Table 5 shows the experimental data. We can see that the worst-case estimate in Lemma 14 is quite inaccurate when ϵ is large, whereas our estimate in Lemma 15 matches the actual case quite well.

Table 5. Experimental measure of expected repetitions. Data is collected from 1000 MITAKA signatures over $\mathcal{R} = \mathbb{Z}[x]/(x^n + 1)$ with $n = 1024$.

	Experimental data	Estimated bound (Lemma 15)	Worst-case bound (Lemma 14)
$\epsilon^{-1} = n$	7.359	7.367	54.598
$\epsilon^{-1} = 2n$	2.715	2.716	7.389
$\epsilon^{-1} = 4n$	1.648	1.648	2.718
$\epsilon^{-1} = 6n$	1.395	1.395	1.947
$\epsilon^{-1} = 8n$	1.284	1.284	1.649
$\epsilon^{-1} = 10n$	1.221	1.221	1.492
$\epsilon^{-1} = 12n$	1.181	1.181	1.396

4.4 Theta Function Division

In the modified samplers (Algorithms 1, 2, 3 and 4), the computation of rejection probability boils down to the division of two theta functions, i.e. $\frac{\rho_{s,c}(\mathbb{Z})}{\rho_s(\mathbb{Z})}$. The term

Algorithm 5: The ThetaDiv algorithm

Input: The center c , the width s and the iteration number k

Output: $\Delta = \frac{\rho_{s,c}(\mathbb{Z})}{\rho_s(\mathbb{Z})}$

```

1  $\Delta \leftarrow 1$ 
2  $r_1 \leftarrow \cos(2\pi c)$ 
3  $r_2 \leftarrow \exp(-\pi s^2)$ 
4 for  $i = 1, \dots, k$  do
5    $\Delta \leftarrow \Delta \cdot \left(1 + \frac{2r_1 - 2}{r_2^{2i-1} + r_2^{1-2i} + 2}\right)$ 
6 end for
7 return  $\Delta$ 

```

$\rho_{s,c}(\mathbb{Z})$ can be computed using the Poisson summation formula as suggested in [BLP⁺13]. In this work, we propose a new algorithm for this calculation in Algorithm 5, which may be of independent interest. A notable advantage of Algorithm 5 is that it separates the center c and the subscript k of the theta series, thus it suffices to only compute $\cos(2\pi c)$ on the fly.

Lemma 16. *Algorithm 5 is correct when $k = +\infty$.*

Proof. Using the Poisson summation formula we get

$$\Delta = \frac{\rho_{s,c}(\mathbb{Z})}{\rho_s(\mathbb{Z})} = \frac{s \cdot \sum_{k \in \mathbb{Z}} \exp(-\pi k^2 s^2) \cdot \exp(2\pi i k c)}{s \cdot \sum_{k \in \mathbb{Z}} \exp(-\pi k^2 s^2)} = \frac{\sum_{k \in \mathbb{Z}} \exp(-\pi k^2 s^2) \cdot \exp(2\pi i k c)}{\sum_{k \in \mathbb{Z}} \exp(-\pi k^2 s^2)}.$$

The numerator can be written in the form of the Jacobi theta function [SS10] as

$$\sum_{k \in \mathbb{Z}} \exp(-\pi k^2 s^2) \cdot \exp(2\pi i k c) = \Theta(c \mid \tau) \quad \text{with } \tau = i s^2.$$

Let $q = \exp(\pi i \tau) = \exp(-\pi s^2)$, then the Jacobi triple-product gives

$$\Theta(c \mid \tau) = \Pi(c \mid \tau) := \prod_{n=1}^{\infty} (1 - q^{2n})(1 + q^{2n-1} \exp(2\pi i c))(1 + q^{2n-1} \exp(-2\pi i c)).$$

Immediately it follows that

$$\begin{aligned}
\Delta &= \frac{\Theta(c \mid \tau)}{\Theta(0 \mid \tau)} = \frac{\Pi(c \mid \tau)}{\Pi(0 \mid \tau)} \\
&= \prod_{n=1}^{\infty} \frac{(1 + q^{2n-1} \exp(2\pi ic))(1 + q^{2n-1} \exp(-2\pi ic))}{(1 + q^{2n-1})^2} \\
&= \prod_{n=1}^{\infty} \left(1 + \frac{q^{2n-1} \exp(2\pi ic) + q^{2n-1} \exp(-2\pi ic) - 2q^{2n-1}}{1 + q^{4n-2} + 2q^{2n-1}} \right) \\
&= \prod_{n=1}^{\infty} \left(1 + \frac{\exp(2\pi ic) + \exp(-2\pi ic) - 2}{q^{1-2n} + q^{2n-1} + 2} \right) \\
&= \prod_{n=1}^{\infty} \left(1 + \frac{2 \cos(2\pi c) - 2}{\exp((2n-1)\pi s^2) + \exp((1-2n)\pi s^2) + 2} \right).
\end{aligned}$$

This proves the lemma. $\square$

Convergence speed. Algorithm 5 converges with k very fast, hence it suffices to set a small k in practical implementations. Lemma 17 gives a theoretical analysis of the convergence rate. For original FALCON parameter $s = 3.2$, $k = 1$ is sufficient to get a result close to the exact value within relative error less than 2^{-64} for any center. For a smaller $s \approx 1.56$ as in Section 5, $k = 2$ already ensures error below 2^{-53} for any center.

Lemma 17 (Convergence rate of ThetaDiv). *For any $k \geq 1$, define*

$$\Delta_k = \prod_{n=1}^k \left(1 + \frac{2 \cos(2\pi c) - 2}{\exp((2n-1)\pi s^2) + \exp(-(2n-1)\pi s^2) + 2} \right).$$

Then, we have

$$0 \leq \frac{\Delta_k}{\Delta} - 1 \leq \exp\left(\frac{4 \exp(-(2k+1)\pi s^2)}{1 - \exp(-2\pi s^2)}\right) - 1.$$

Proof. For notational convenience, we denote

$$\frac{\Delta_k}{\Delta} = \prod_{n=k+1}^{\infty} (1 + a_n)^{-1}, \quad a_n := \frac{2 \cos(2\pi c) - 2}{\exp((2n-1)\pi s^2) + \exp(-(2n-1)\pi s^2) + 2}.$$

Since $2 \cos(2\pi c) - 2 \in [-4, 0]$, we have $a_n \in [-1, 0]$ and

$$|a_n| \leq 4 \exp(-(2n-1)\pi s^2).$$

Using $|\log(1+x)| \leq |x|$ for $x \in [-1, 0]$, we obtain

$$\log \frac{\Delta_k}{\Delta} = - \sum_{n=k+1}^{\infty} \log(1 + a_n) \leq \sum_{n=k+1}^{\infty} |a_n| \leq \frac{4 \exp(-(2k+1)\pi s^2)}{1 - \exp(-2\pi s^2)}.$$

Exponentiating both sides yields

$$\frac{\Delta_k}{\Delta} \leq \exp\left(\frac{4 \exp(-(2k+1)\pi s^2)}{1 - \exp(-2\pi s^2)}\right),$$

which proves the lemma. $\square$

Efficiency. We test the running time of Algorithm 5 on an Intel Core i5-1135G7 CPU clocked at 2.4 GHz. The inverse of $(r_2^{2i-1} + r_2^{1-2i} + 2)$ can be precomputed along with s during key generation. For the parameters of Falcon-512, the cycle count for calling Algorithm 5 1024 times is around 1.2 Mcycles, whereas the cycle count for the FALCON signing process is 7.3 Mcycles. Therefore Algorithm 5 would only cause a mild overhead on the efficiency.

5 Practical NTRU-Based Signatures with Weak Smoothness

5.1 FALCON^{ws}: A Compact Variant of FALCON

In this section, we present a variant of FALCON using weak smoothness condition. Compared to Falcon, our scheme achieves higher security and its signature size is smaller by more than 25%. Our scheme uses a modified FALCON sampler to work with weak smoothness. Since the FALCON sampler, known as the fast Fourier sampler [DP16], is a ring-efficient variant of Klein-GPV sampler, its adaption to weak smoothness can be done in the same manner with Algorithm 1, which we do not demonstrate. The rest of this section will mainly focus on the concrete efficiency of our scheme.

5.1.1 Tight Estimate of Rejection Rate. Algorithm 1 uses rejecting sampling to ensure the correct output, which yields an efficiency loss. Lemma 11 gives an upper bound of the average trial number, but that bound is too overestimated in the context of FALCON. This is because $\overline{\eta}_\epsilon(\mathbb{Z})$ is the lower bound of all σ_i 's, however in FALCON the first and second half of σ_i 's are gradually decreasing and $\max_i\{\sigma_i\} \approx \sqrt{2} \min_i\{\sigma_i\}$. We conduct a tighter analysis of the rejection rate of our FALCON variant based on the following heuristics:

- *Heuristic 1:* $\|\tilde{\mathbf{b}}_{n+i}\| = \|\tilde{\mathbf{b}}_i\|$ for $0 \leq i < n$;
- *Heuristic 2:* $\|\tilde{\mathbf{b}}_{2i}\| = \|\tilde{\mathbf{b}}_{2i+1}\| = \sqrt{\frac{n-i}{n}} \|\tilde{\mathbf{b}}_0\|$ for $0 \leq i < \frac{n}{2}$.

The first heuristic stems from that FALCON restricts $\|\tilde{\mathbf{b}}_0\| \approx \|\tilde{\mathbf{b}}_n\| \approx 1.17\sqrt{q}$ to get a compact trapdoor and that the orthogonalization on the last n vectors is basically the same as on the first n vectors. The property of $\|\tilde{\mathbf{b}}_{2i}\| = \|\tilde{\mathbf{b}}_{2i+1}\|$ holds in the fast Fourier orthogonalization [DP16]. The second heuristic follows from the geometric assumption. Similar heuristics were used in [Pre15, Section 6.5.1] to explain the optimal parameter for NTRU trapdoors. Lemma 18 shows our tight estimate of the repetition number in our FALCON variant.

Lemma 18. *Under above heuristics, the approximate upper bound of the repetition number of the modified FALCON sampler for $0 < \epsilon \leq 0.047$ is*

$$\exp\left(8.4 \cdot \left(-\frac{n\epsilon^2}{8} + \frac{n\epsilon}{2} + n \log\left(\frac{\epsilon}{2}\right) \cdot \left(Li\left(\frac{\epsilon^2}{4}\right) - Li\left(\frac{\epsilon}{2}\right)\right)\right)\right)$$

where $Li(x)$ is the logarithmic integral function.

To prove Lemma 18, we need the following simple lemma.

Lemma 19. *For any $c > 1$, if $s \geq \eta_\epsilon(\mathbb{Z})$, we have $\sqrt{cs} \geq \eta_{(\frac{\epsilon}{2})^c}(\mathbb{Z})$.*

Proof. According to the definition of smoothing parameter, it holds that

$$\rho_{\frac{1}{s}}(\mathbb{Z}) = 1 + 2 \cdot \sum_{k \in \mathbb{Z}^+} e^{-\pi s^2 k^2} \leq 1 + \epsilon.$$

and then

$$\rho_{\frac{1}{\sqrt{cs}}}(\mathbb{Z}) = 1 + 2 \cdot \sum_{k \in \mathbb{Z}^+} e^{-\pi cs^2 k^2} \leq 1 + 2 \cdot \left(\sum_{k \in \mathbb{Z}^+} e^{-\pi s^2 k^2}\right)^c \leq 1 + 2 \left(\frac{\epsilon}{2}\right)^c.$$

The proof is completed. $\square$

Proof of Lemma 18. By Lemma 1 and $s_i = \frac{s}{\|\mathbf{b}_i\|}$, we have

$$M = \prod_{i=0}^{2n-1} \frac{\rho_{s_i}(\mathbb{Z})}{\rho_{s_i, c_i''}(\mathbb{Z})} = \prod_{i=0}^{2n-1} \frac{\rho_{s/\|\mathbf{b}_i\|}(\mathbb{Z})}{\rho_{s/\|\mathbf{b}_i\|, c_i''}(\mathbb{Z})} \leq \prod_{i=0}^{2n-1} \frac{1 + \epsilon_i}{1 - \epsilon_i}$$

where ϵ_i such that $s_i = \eta_{\epsilon_i}(\mathbb{Z})$. Then $\epsilon_i = \epsilon_{i+n}$ and particularly $\epsilon_0 = \epsilon_n = \epsilon$ under Heuristic 1. By Heuristic 2 and Lemma 19, we further have

$$M \leq \left(\prod_{i=0}^{\frac{n}{2}-1} \frac{1 + \epsilon_{2i}}{1 - \epsilon_{2i}}\right)^4 = \left(\prod_{i=0}^{\frac{n}{2}-1} 1 + \frac{2\epsilon_{2i}}{1 - \epsilon_{2i}}\right)^4 \leq \exp\left(8 \cdot \sum_{i=0}^{\frac{n}{2}-1} \frac{\epsilon_{2i}}{1 - \epsilon_{2i}}\right).$$

and $\epsilon_{2i} = \left(\frac{\epsilon}{2}\right)^{\frac{n}{n-i}} \leq \epsilon$. The function $f(x) = \frac{x}{1-x} < 1.05x$ when $0 < x \leq 0.047$. Putting these together, we get

$$\begin{aligned} M &\leq \exp\left(8.4 \cdot \sum_{i=0}^{\frac{n}{2}-1} \epsilon_{2i}\right) = \exp\left(8.4 \cdot \sum_{i=0}^{\frac{n}{2}-1} \left(\frac{\epsilon}{2}\right)^{\frac{n}{n-i}}\right) \\ &\approx \exp\left(8.4 \cdot \int_0^{\frac{n}{2}} \left(\frac{\epsilon}{2}\right)^{\frac{n}{n-x}} dx\right) \\ &= \exp\left(8.4 \cdot \left(-\frac{n\epsilon^2}{8} + \frac{n\epsilon}{2} + n \log\left(\frac{\epsilon}{2}\right) \cdot \left(Li\left(\frac{\epsilon^2}{4}\right) - Li\left(\frac{\epsilon}{2}\right)\right)\right)\right) \end{aligned}$$

The proof is finished. $\square$

Table 6 shows the experimental data measured over random FALCON bases. It can be seen that the worst-case estimate in Lemma 11 is quite inaccurate when ϵ is large, whereas our estimate in Lemma 18 matches the actual case well.

Table 6. Experimental measure of expected repetitions. Data is collected from 100 random FALCON basis over $\mathcal{R} = \mathbb{Z}[x]/(x^n + 1)$ with $n = 512, 1024$. The item “ $A/B/C$ ” represents the actual mean of average repetition, the estimate of Lemma 18 and the worst-case bound of Lemma 11.

	$\epsilon^{-1} = n$	$\epsilon^{-1} = 2n$	$\epsilon^{-1} = 4n$
$n = 512$	1.48 / 1.62 / 54.6	1.20 / 1.25 / 7.4	1.09 / 1.08 / 2.7
$n = 1024$	1.43 / 1.56 / 54.6	1.18 / 1.23 / 7.4	1.08 / 1.1 / 2.7

5.1.2 Parameter. We choose $\epsilon = \frac{1}{2n}$ for FALCON^{ws}. As per Lemma 18, the expected number of repetitions is less than 1.25 for the modified FALCON-512 and FALCON-1024. This leads to a moderate efficiency loss. Interestingly, the weak smoothness allows our scheme to use a smaller modulus q , which reduces both key and signature sizes. Indeed reducing the modulus requires particular care: once q is reduced to a value much smaller than the signature length as in [ETWY22], it opens the avenue to the Z-shape attack [DEP23]. However, the weak smoothness substantially reduces the signature Gaussian width and thus amplifies the ratio between the modulus and the signature length, which admits a smaller and safe modulus.

The detailed parameters of our scheme are shown in Table 7. Compared to FALCON, our scheme has both higher security and smaller key and signature sizes. By the way, the parameters chosen all satisfy the requirements of Lemma 10 to ensure the Rényi Divergence is small enough even with 2^{64} signatures. Specifically, for $n = 512$, FALCON^{ws} chooses an $q = 953$ and its signature (resp. public key) size is smaller by 29% (resp. 28%) than FALCON-512 while its security is higher by 21 bits. For $n = 1024$, we propose two parameter sets achieving the security level NIST-5. In the parameter set for small size, $q = 1949$ and compared to FALCON-1024, the signature size is reduced by 23.5% and the public key size is reduced by 21% while its security is higher by 10 bits. The parameter set for higher security sets an NTT-compatible $q = 3329$ to allow a balance between key recovery security and forgery security by increasing the secret key size. Its signature and key sizes are still smaller by 16% and 14% than Falcon-1024 while its security is higher by 30 bits.

The signature size is estimated as per the entropic bound which can be closely achieved by the rANS encoding in Section 7.

Table 7. Parameter sets for FALCON^{ws}.

Target NIST Security Level	1	5	5+
Ring degree n	512	1024	1024
Modulus q	953	1949	3329
Signature Gaussian std σ	26.49	39.1	63.67
Secret coefficient std $\sigma_{f,g}$	1.33	1.33	2.17
Repetitions(with $\epsilon = \frac{1}{2n}$ for $\eta_\epsilon(\mathbb{Z})$)	1.25	1.23	1.23
Max. signature norm $\beta = \lceil 1.1 \cdot \sigma\sqrt{2n} \rceil$	933	1946	3170
Public key bytelength	640	1408	1536
Signature bytelength	474	979	1070
Key-recovery attack: BKZ B/C/Q	485/142/129	973/284/258	1037/303/275
Forgery attack: BKZ B/C/Q	515/150/136	1139/333/302	1058/309/280

5.2 NTRU-based Signatures using the Modified Peikert Sampler

This section validates the practicality of GPV signatures using Peikert sampler by showing an NTRU trapdoor based instantiation. Our scheme offers a simpler and fully parallelizable implementation compared to FALCON and MITAKA [EFG⁺22]. Additionally, it would be easier to implement without using floating-point arithmetic [DGPY20]. Due to the inherent quality limitation of Peikert sampler, the overall performance of our scheme is worse than FALCON and MITAKA, whereas its signature size is smaller than Dilithium-3 by about 49% for the security level NIST-3.

5.2.1 Optimal NTRU Trapdoor for Peikert Sampler. The quality $Q(\mathbf{B}_{f,g})$ in Peikert sampler is the largest singular value $s_1(\mathbf{B}_{f,g})$. As shown in [Pre15, Table 6.1], the optimal $s_1(\mathbf{B}_{f,g})$ for a random NTRU trapdoor is about $8\sqrt{q}$ with dimension $2n = 1024$. This results in a significant security loss even with weak smoothness. Inspired by MITAKA [EFG⁺22] and ANTRAG [ENS⁺23], we attempt to refine the trapdoor generation for Peikert sampler to mitigate the security loss.

By [Pre15, Lemma 6.8], it holds that

$$s_1(\mathbf{B}_{f,g}) = \frac{1}{2} \sqrt{\max_i (\varphi_i(T + \sqrt{T^2 - 4q^2}))}$$

where $T = ff^* + gg^* + FF^* + GG^*$. We note that $s_1(\mathbf{B}_{f,g})$ is determined by $\max_i \varphi_i(T)$. Given the expensive cost of solving NTRU equation, we would like to be able to determine when $\max_i \varphi_i(T)$ tends to be small enough only by (f, g) . By the unique of $\tilde{F}, \tilde{G}$ described in Lemma 7, T can be written as

$$T = (1 + rr^*)(ff^* + gg^*) + \frac{q^2}{ff^* + gg^*}$$

Algorithm 6: The Peikert sampler NTRU trapdoor generation

Input: A degree n , a modulus q , a target quality τ
Output: The NTRU trapdoor $\mathbf{B}_{f,g}$ such that $s_1(\mathbf{B}_{f,g}) \leq \tau\sqrt{q}$.

```

1 repeat
2   repeat
3     for  $1 \leq i \leq n/2$  do
4       sample  $(z_i, w_i) \in \mathbb{C}^2$  uniformly such that
          $(|z_i|, |w_i|) \in A^+(\sqrt{r}q, \sqrt{R}q)$ 
5     end for
6      $\hat{f} \leftarrow \varphi^{-1}(z_1, \dots, z_{d/2}) \in \mathcal{K}_{\mathbb{R}}$ 
7      $\hat{g} \leftarrow \varphi^{-1}(w_1, \dots, w_{d/2}) \in \mathcal{K}_{\mathbb{R}}$ 
8      $f \leftarrow \lfloor \hat{f} \rfloor, g \leftarrow \lfloor \hat{g} \rfloor$ 
9   until  $\alpha q \leq \varphi_i(ff^* + gg^*) \leq \beta q$  for all  $i = 1, \dots, n/2$ 
10   $\mathbf{B}_{f,g} \leftarrow \text{NTRUSolver}(f, g, q)$ 
11 until  $s_1(\mathbf{B}_{f,g}) \leq \tau\sqrt{q}$ 
12 return  $\mathbf{B}_{f,g}$ 

```

where $F = \tilde{F} - rf$, $G = \tilde{G} - rg$ and $r = \frac{Ff^* + Gg^*}{ff^* + gg^*}$. Suppose that the coefficients of r are uniform over $[-\frac{1}{2}, \frac{1}{2})$, then $\max_i \varphi_i(T)$ is expected to be minimized when

$$\max_i \varphi_i((1 + rr^*)(ff^* + gg^*)) \approx \max_j \varphi_j\left(\frac{q^2}{ff^* + gg^*}\right).$$

To achieve the optimum, we apply the technique in [ENS⁺23] by restricting (f, g) such that $\alpha q \leq \varphi_i(ff^* + gg^*) \leq \beta q$. Detailed algorithm is shown in Algorithm 6. We experimentally evaluate how small $s_1(\mathbf{B}_{f,g})$ can be. In order to avoid too many rejections during the generation of (f, g) , the parameters r, R, α, β will change slightly with q . However, τ is basically determined by the dimension n and is independent of q . Finally we obtain

- for $(n, q) = (512, 20000)$, $(r, R) = (\frac{1}{14}, \frac{1}{12})$ and $(\alpha, \beta) = (\frac{1}{25}, \frac{3}{20})$ give a nearly optimal quality $s_1(\mathbf{B}_{f,g}) \approx 6\sqrt{q}$;
- for $(n, q) = (1024, 80000)$, $(r, R) = (\frac{1}{19}, \frac{1}{17})$ and $(\alpha, \beta) = (\frac{3}{100}, \frac{1}{10})$ give a nearly optimal quality $s_1(\mathbf{B}_{f,g}) \approx 8\sqrt{q}$.

5.2.2 Parameter. Table 8 presents two parameter sets for our scheme that integrates the modified Peikert sampler (Algorithms 2 and 3) and the refined NTRU trapdoor generation (Algorithm 6). Our security estimate has considered the subfield attack discussed in [ENS⁺23] that exploits the additional geometric information that all embeddings of (f, g) lie in an annulus. It turns out that our parameters are not vulnerable to this attack. While our scheme using modified Peikert sampler is less efficient than FALCON and MITAKA, its signature size is

practical and even smaller by around 49% compared to Dilithium for the NIST security level-3.

Table 8. Parameter sets for our scheme using modified Peikert sampler

Ring degree n	512	1024
Modulus q	20000	80000
Signature Gaussian std σ	580.36	1604.66
Repetitions(with $\epsilon = \frac{1}{4}$ for $\eta_\epsilon(\mathbb{Z}^{2n})$)	1.67	1.67
Max. signature norm $\beta = \lceil 1.05 \cdot \sigma\sqrt{2n} \rceil$	19501	76250
Public key bytelength	960	2176
Signature bytelength	759	1655
Key-recovery attack: BKZ B/C/Q	306/89/81	646/189/171
Forgery attack: BKZ B/C/Q	302/88/80	677/198/179

5.3 ANTRAG with Weak Smoothness

5.3.1 ANTRAG^{ws}: A Compact Variant of ANTRAG. This section presents the ANTRAG^{ws} signature scheme that is the adaption of ANTRAG to weak smoothness using Algorithm 4 with $d = k = 2$. We fix $\epsilon = \frac{1}{8n}$ for $r = \overline{\eta}_\epsilon(\mathbb{Z})$. By Lemma 15, the expected number of repetitions is less than 1.28, which results in only a moderate efficiency loss.

The weak smoothness condition enables the use of a smaller modulus q . We choose q such that the maximum signature norm remains strictly smaller than the modulus q , which completely defeats the Z-shape attack [DEP23]. As observed in [ENS⁺23], the choice of q affects the quality of the generated trapdoor (f, g) : for a fixed number of key generation trials, a smaller q tends to yield a larger trapdoor quality parameter α for all i ,

$$q/\alpha^2 \leq |\varphi_i(f)|^2 + |\varphi_i(g)|^2 < \alpha^2 q.$$

For a fair comparison, we follow the reference implementation of ANTRAG¹ and evaluate the trapdoor quality under our reduced modulus q , while keeping 3 – 5 key generation trials in average. The resulting α is larger than that in ANTRAG with $q = 12289$, which is shown in Table 9.

We further evaluate the bit security against key recovery attacks by analyzing $\|(f, g)\|$. We experimentally determine γ such that

$$\Pr [\|(f, g)\| < \gamma\sqrt{q}] \leq 1\%$$

¹ <https://github.com/mti/antrag>

and reject (f, g) of norm less than $\gamma\sqrt{q}$. This has a minor impact on the overall key generation cost and allows us to use $\gamma\sqrt{q}$ as a conservative lower bound in the security analysis.

The detailed parameters of ANTRAG^{ws} are summarized in Table 9. For $n = 512$, ANTRAG^{ws} chooses an $q = 1237$ and its signature (resp. public key) size is reduced by 21.8% (resp. 21.4%) compared with ANTRAG-512. For $n = 1024$, ANTRAG^{ws} chooses an $q = 2729$ and its signature (resp. public key) size is reduced by 16.5% (resp. 14.3%) compared with ANTRAG-1024.

Table 9. Parameter sets for ANTRAG^{ws}.

Target NIST Security Level	1	5
Ring degree n	512	1024
Modulus q	1237	2729
Trapdoor quality (α, γ)	(1.55, 1.11)	(1.58, 1.13)
Signature Gaussian std σ	36.83	57.87
Repetitions(with $\epsilon = \frac{1}{8n}$ for $\eta_\epsilon(\mathbb{Z})$)	1.28	1.28
Max. signature norm $\beta = \lceil 1.1 \cdot \sigma\sqrt{2n} \rceil$	1234	2705
Public key bytelength	704	1536
Signature bytelength	505	1052
Key-recovery attack: BKZ B/C/Q	451/131/119	927/270/245
Forgery attack: BKZ B/C/Q	485/141/128	1079/315/285

5.3.2 ANTRAG^{ws} without Floating-Point Arithmetic. Floating-point arithmetic (FPA) could introduce additional implementation complications and security risks. Along with MITAKA, Espitau et al. proposed an FPA-free variant MITAKA _{$\mathbb{Z}$} achieving the same size and security [EFG⁺22]. The integer arithmetic friendly sampler of MITAKA _{$\mathbb{Z}$} can be directly used in ANTRAG^{ws}, but it comes with greatly more base samplings.

We present a new FPA-free hybrid sampler and use it to remove FPA from ANTRAG^{ws}. Our main idea is to use an approximate square Gram root decomposition to avoid the overly wide integral Gram decomposition [DGPY20] and then to rectify the discrepancy with rejection sampling again, which is inspired by [Duc14, Algorithm 28]. To address the floating-point arithmetic in the added rejection sampling, we approximate the inverse of the covariance matrix using the expansion $(\mathbf{I} - \Sigma_\delta)^{-1} = \sum_{i=0}^{\infty} \Sigma_\delta^i$, and truncate the series after a few terms. In practice, one or two terms already suffice to meet the precision requirement. The detailed algorithm is given in Algorithm 7. The matrix decomposition in Algorithm 7 is computed using a ring-based Cholesky decomposition, preserving the ring structure of the resulting matrix $\mathbf{A}$. This allows to implement the sampler in a *quasi-linear* time.

Algorithm 7: The FPA-free hybrid sampler

Input: Basis $\hat{\mathbf{B}} \in \frac{1}{p}\mathcal{R}^{2 \times 2}$ such that $\hat{\mathbf{B}}\mathbf{U}_{\hat{u}} = \mathbf{B} = \tilde{\mathbf{B}}\mathbf{U}_u$ where $\hat{u} = [u]_p \in \frac{1}{p}\mathcal{R}$, a center $\mathbf{c} \in \mathcal{R}^2$ and parameter $r, s > 0$.
Output: A lattice point $\mathbf{z}$ following a distribution close to $D_{\mathcal{L}(\mathbf{B}), sr, \mathbf{c}}$.

- 1 Precomputation: $\Sigma_p = s^2\mathbf{I} - \hat{\mathbf{B}}\hat{\mathbf{B}}^t$ and $\hat{\mathbf{A}} = [\mathbf{A}]_p$ such that $\mathbf{A}\mathbf{A}^t = \Sigma_p - \mathbf{I}$ and $\Sigma_\delta = \Sigma_p - \mathbf{I} - \hat{\mathbf{A}}\hat{\mathbf{A}}^t$, $\mathbf{T} = \sum_{i=1}^k (\Sigma_\delta/s^2)^i$
- 2 $\mathbf{p} \leftarrow \text{PerturbationSamp}(\hat{\mathbf{A}})$
- 3 $\hat{\mathbf{c}} = \hat{\mathbf{B}}^{-1}(\mathbf{c} - \mathbf{p})$ /* $\hat{\mathbf{c}} = (\hat{c}_1, \hat{c}_2)$ */
- 4 $z_2 \leftarrow \text{RingSampler}_{\mathbb{Z}}(\hat{c}_2, r)$
- 5 $c'_1 \leftarrow \hat{c}_1 - z_2\hat{u}$
- 6 $z_1 \leftarrow \text{RingSampler}_{\mathbb{Z}}(c'_1, r)$ /* $\mathbf{c}' = (\hat{c}_2, c'_1)$ */
- 7 $\mathbf{z} \leftarrow \mathbf{B}(z_1, z_2)$ with prob. $\Delta_1 = \frac{\rho_{r, \mathbf{c}'}(\mathcal{R}^2)}{\rho_r(\mathcal{R}^2)}$ otherwise restart
- 8 **if** $\|\mathbf{z} - \mathbf{c}\| > (1 + \tau)\sqrt{(s^2 + \delta)r\sqrt{n/\pi}}$ **then** restart
- 9 **return** $\mathbf{z}$ with prob. $\Delta_2 = \frac{1}{C} \cdot \exp\left(\pi \frac{(\mathbf{z} - \mathbf{c})^t \mathbf{T}(\mathbf{z} - \mathbf{c})}{r^2 s^2}\right)$ otherwise restart

Algorithm 8: PerturbationSamp

Input: A matrix $\hat{\mathbf{A}} \in \frac{1}{p}\mathcal{R}^{2 \times 2}$.
Output: $\mathbf{p} \in \frac{1}{L}\mathcal{R}^2$ with distribution like $D_{\frac{1}{L}\mathcal{R}^2, r\sqrt{\Sigma_p'}}$, where $\Sigma_p' = \hat{\mathbf{A}}\hat{\mathbf{A}}^t + \mathbf{I} = \Sigma_p - \Sigma_\delta$.

- 1 Precompute: $L \in \mathbb{Z}$ such that $Lr > \eta_{\epsilon'}(\mathcal{R}^2)$ and $Lr\sqrt{\mathbf{I} - \Sigma_p'^{-1}} \geq \eta_{\epsilon'}(\mathcal{R}^2)$
- 2 $\mathbf{x} \leftarrow ([0]_{Lr})^m$
- 3 $\mathbf{p}' \leftarrow \hat{\mathbf{A}}\mathbf{x}$
- 4 $\mathbf{p} \leftarrow \frac{1}{L} \lfloor \mathbf{p}' \rfloor_{Lr}$
- 5 **return** $\mathbf{p}$

Lemma 20. Keep the notation of Algorithm 7. Let $\epsilon' > 0$ be negligible, $s \in \mathbb{Z}$ such that

$$s > |\mathbf{B}_{f,g}|_{\mathcal{K}} \left(1 + \sqrt{2n/p}\right) + 1,$$

and $r \geq \eta_{\epsilon}(\mathbb{Z})$. Define

$$\Sigma'^{-1} = \left(\hat{\mathbf{B}}\hat{\mathbf{B}}^t\right)^{-1} + (\Sigma_p')^{-1}.$$

and choose an integer L such that

$$Lr\Sigma' \geq \eta_{\epsilon'}(\mathcal{R}^2), \quad Lr \geq \eta_{\epsilon'}(\mathcal{R}^2), \quad Lr\sqrt{\mathbf{I} - \Sigma_p'^{-1}} \geq \eta_{\epsilon'}(\mathcal{R}^2).$$

Let p and k be positive integers, and $\tau > 0$ such that the following conditions hold for all $\mathbf{c} \in \mathcal{R}^2$:

$$\frac{n}{2p} \sqrt{s_1 (\boldsymbol{\Sigma}_p - \mathbf{I})} + \frac{n^2}{p^2} \leq \delta$$

and

$$(1 + \tau)^2 \cdot n \cdot \frac{s^2 + \delta}{s^2 - \delta} \cdot \left(\frac{\delta}{s^2} \right)^{k+1} \leq \epsilon'/2$$

and

$$\frac{\rho_{r\sqrt{s^2+\delta}}(\mathcal{L}(\mathbf{B}))}{\rho_{r\sqrt{s^2-\delta}}(\mathcal{L}(\mathbf{B}) - \mathbf{c})} \cdot \exp(-n\tau^2) \leq \epsilon'.$$

Let $C \in \frac{1}{p}\mathbb{Z}$ such that

$$C > \exp\left(\frac{(1 + \tau)^2(s^2 + \delta)n\delta}{(s^2 - \delta)s^2}\right).$$

Then the Rényi divergence between the output distribution of Algorithm 7 and $D_{\mathcal{L}(\mathbf{B}), \mathbf{c}, sr}$ is at most

$$\frac{(1 + \epsilon')^5}{(1 - \epsilon')^3}.$$

Moreover, the expected number of repetitions is at most

$$\frac{C^2(1 + \epsilon)^{2n}}{(1 - \epsilon)^{2n}}.$$

Proof. We begin with an analysis of the output distribution of **PerturbationSamp**. Here $\boldsymbol{\Sigma}'_p$ denotes the intermediate covariance matrix arising from the offline sampling step, and $\boldsymbol{\Sigma}_\delta$ captures the approximation error such that $\boldsymbol{\Sigma}_p = \boldsymbol{\Sigma}'_p + \boldsymbol{\Sigma}_\delta$.

The distribution of PerturbationSamp. We show that the vector $\mathbf{p}$ satisfies

$$\begin{aligned} P(\mathbf{p}) &= \sum_{\mathbf{p}' \in \mathcal{R}^2} \frac{\rho_{Lr\sqrt{\boldsymbol{\Sigma}'_p - \mathbf{I}}}(\mathbf{p}')}{\rho_{Lr}(\mathcal{R}^2)} \cdot \frac{\rho_{Lr}(L\mathbf{p} - \mathbf{p}')}{\rho_{Lr, \mathbf{p}'}(\mathcal{R}^2)} \\ &= \rho_{r\sqrt{\boldsymbol{\Sigma}'_p}}(\mathbf{p}) \cdot \sum_{\mathbf{p}' \in \mathcal{R}^2} \frac{\rho_{Lr\sqrt{\mathbf{I} - \boldsymbol{\Sigma}'_p^{-1}}}(\mathbf{p}' - \mathbf{c}_{p'})}{\rho_{Lr}(\mathcal{R}^2)\rho_{Lr, \mathbf{p}'}(\mathcal{R}^2)} \\ &:= \rho_{r\sqrt{\boldsymbol{\Sigma}_p - \boldsymbol{\Sigma}_\delta}}(\mathbf{p}) \cdot C_{\mathbf{p}}. \end{aligned}$$

Since $Lr > \eta_{\epsilon'}(\mathcal{R}^2)$ and $Lr\sqrt{\mathbf{I} - \boldsymbol{\Sigma}'_p{}^{-1}} \geq \eta_{\epsilon'}(\mathcal{R}^2)$, Lemma 1 implies that

$$C_{\mathbf{p}} \in \left[\left(\frac{1 - \epsilon'}{1 + \epsilon'} \right), \left(\frac{1 + \epsilon'}{1 - \epsilon'} \right) \right] \cdot \frac{\rho_{Lr\sqrt{\mathbf{I} - \boldsymbol{\Sigma}'_p{}^{-1}}}(\mathcal{R}^2)}{\rho_{Lr}(\mathcal{R}^2)\rho_{Lr}(\mathcal{R}^2)}.$$

Distribution of $\mathbf{z}$ in Step 7 (after rejection). Denote by $\mathbf{z}' = \mathbf{U}_{\hat{u}}(z_1, z_2)$. We now turn to the distribution of $\mathbf{z}$ in Step 7:

$$\begin{aligned}
P(\mathbf{z}) &= \sum_{\mathbf{p} \in \frac{1}{L}\mathcal{R}^2} C_{\mathbf{p}} \cdot \rho_{r\sqrt{\Sigma_p - \Sigma_\delta}}(\mathbf{p}) \cdot \frac{\rho_{r,c_2}(z_2)}{\rho_{r,c_2}(\mathcal{R})} \cdot \frac{\rho_{r,c'_1}(z_1)}{\rho_{r,c'_1}(\mathcal{R})} \cdot \frac{\rho_{r,\mathbf{c}'}(\mathcal{R}^2)}{\rho_r(\mathcal{R}^2)} \\
&= \sum_{\mathbf{p} \in \frac{1}{L}\mathcal{R}^2} C_{\mathbf{p}} \cdot \rho_{r\sqrt{\Sigma_p - \Sigma_\delta}}(\mathbf{p}) \cdot \frac{\rho_{r,\hat{\mathbf{c}}}(\mathbf{z}')}{\rho_{r,\mathbf{c}'}(\mathcal{R}^2)} \cdot \frac{\rho_{r,\mathbf{c}'}(\mathcal{R}^2)}{\rho_r(\mathcal{R}^2)} \\
&= \sum_{\mathbf{p} \in \frac{1}{L}\mathcal{R}^2} \frac{C_{\mathbf{p}}}{\rho_r(\mathcal{R}^2)} \cdot \rho_{r\sqrt{\Sigma_p - \Sigma_\delta}}(\mathbf{p}) \cdot \rho_{r\hat{\mathbf{B}}}(\mathbf{z} - (\mathbf{c} - \mathbf{p})) \\
&= \rho_{r\sqrt{s^2 - \Sigma_\delta}, \mathbf{c}}(\mathbf{z}) \cdot \sum_{\mathbf{p} \in \frac{1}{L}\mathcal{R}^2} \frac{C_{\mathbf{p}} \cdot \rho_{r\sqrt{\Sigma'}}(\mathbf{p} - \mathbf{c}_p)}{\rho_r(\mathcal{R}^2)} \\
&\propto \left[\left(\frac{1 - \epsilon'}{1 + \epsilon'} \right), \left(\frac{1 + \epsilon'}{1 - \epsilon'} \right) \right] \cdot \rho_{r\sqrt{s^2 - \Sigma_\delta}, \mathbf{c}}(\mathbf{z}) \cdot \sum_{\mathbf{p} \in \frac{1}{L}\mathcal{R}^2} \rho_{r\sqrt{\Sigma'}}(\mathbf{p} - \mathbf{c}_p) \\
&\propto \left[\left(\frac{1 - \epsilon'}{1 + \epsilon'} \right)^2, \left(\frac{1 + \epsilon'}{1 - \epsilon'} \right) \right] \cdot \rho_{r\sqrt{s^2 - \Sigma_\delta}, \mathbf{c}}(\mathbf{z}).
\end{aligned}$$

The final step follows from the condition $r\sqrt{\Sigma'} \geq \frac{1}{L} \cdot \eta_{\epsilon'}(\mathcal{R}^2)$ together with Lemma 1. Hence the distribution of $\mathbf{z}$ is statistically close to $D_{\mathcal{L}(\mathbf{B}), r\sqrt{s^2 - \Sigma_\delta}, \mathbf{c}}(\mathbf{z})$.

Restart probability and tail bound. We now analyze the restart probability in Step 8. The restart occurs when the sampled vector $\mathbf{z}$ exceeds the prescribed norm bound. The restart probability is as follows:

$$\begin{aligned}
&\Pr \left[\|\mathbf{z} - \mathbf{c}\| > (1 + \tau) \sqrt{(s^2 + \delta)} r \sqrt{\frac{n}{\pi}} \right] \\
&\leq \Pr \left[(\mathbf{z} - \mathbf{c})^t (r^2 s^2 - r^2 \Sigma_\delta)^{-1} (\mathbf{z} - \mathbf{c}) > \frac{(1 + \tau)^2 n}{\pi} \right] \\
&\leq \frac{\rho_{r\sqrt{s^2 - \Sigma_\delta}}(\mathcal{L}(\mathbf{B}))}{\rho_{r\sqrt{s^2 - \Sigma_\delta}}(\mathcal{L}(\mathbf{B} - \mathbf{c}))} \exp(-n\tau^2) \\
&\leq \frac{\rho_{r\sqrt{s^2 + \delta}}(\mathcal{L}(\mathbf{B}))}{\rho_{r\sqrt{s^2 - \delta}}(\mathcal{L}(\mathbf{B} - \mathbf{c}))} \exp(-n\tau^2) \\
&\leq \epsilon'.
\end{aligned}$$

The second inequality follows from Lemma 4 and the third one follows from an upper bound on the operator norm of Σ_δ , which we establish below: for $\mathbf{A}_\delta = \mathbf{A} - \hat{\mathbf{A}}$,

$$\|\Sigma_\delta\| \leq \|\mathbf{A}_\delta \mathbf{A}^t\| + \|\mathbf{A} \mathbf{A}_\delta^t\| + \|\mathbf{A}_\delta \mathbf{A}_\delta^t\| \leq \frac{2n}{p} \sqrt{s_1(\Sigma_p - \mathbf{I})} + \frac{n^2}{p^2} \leq \delta.$$

As a consequence, for the support set

$$\text{Supp} = \{\mathbf{z} : \|\mathbf{z} - \mathbf{c}\| \leq (1 + \tau)\sqrt{(s^2 + \delta)r\sqrt{n/\pi}}\},$$

we have

$$P(\mathbf{z}) \propto \left[\left(\frac{1 - \epsilon'}{1 + \epsilon'} \right)^2, \left(\frac{1 + \epsilon'}{1 - \epsilon'} \right) \right] \cdot \rho_{r\sqrt{s^2 - \Sigma_\delta}, \mathbf{c}}(\mathbf{z}).$$

Final rejection sampling. We now refine the approximation of the distribution of $\mathbf{z}$ by rewriting the density in a form suitable for comparison with $D_{\mathcal{L}(\mathbf{B}), rs, \mathbf{c}}$. Let

$$\mathbf{T} = \sum_{i=1}^k \left(\frac{\Sigma_\delta}{s^2} \right)^i \quad \text{and} \quad \mathbf{T}' = \sum_{i=k+1}^{\infty} \left(\frac{\Sigma_\delta}{s^2} \right)^i.$$

Then

$$\mathbf{I} - \left(\mathbf{I} - \frac{\Sigma_\delta}{s^2} \right)^{-1} = - \sum_{i=1}^{\infty} \left(\frac{\Sigma_\delta}{s^2} \right)^i = -\mathbf{T} - \mathbf{T}'.$$

Using this decomposition, the output distribution of $\mathbf{z}$ can be written as

$$\begin{aligned} P(\mathbf{z}) &\propto \left[\left(\frac{1 - \epsilon'}{1 + \epsilon'} \right)^2, \left(\frac{1 + \epsilon'}{1 - \epsilon'} \right) \right] \cdot \rho_{r\sqrt{s^2 - \Sigma_\delta}, \mathbf{c}}(\mathbf{z}) \cdot \exp \left(\pi \frac{(\mathbf{z} - \mathbf{c})^t \mathbf{T} (\mathbf{z} - \mathbf{c})}{r^2 s^2} \right) \\ &= \left[\left(\frac{1 - \epsilon'}{1 + \epsilon'} \right)^2, \left(\frac{1 + \epsilon'}{1 - \epsilon'} \right) \right] \cdot \rho_{rs, \mathbf{c}}(\mathbf{z}) \cdot \exp \left(-\pi \frac{(\mathbf{z} - \mathbf{c})^t \mathbf{T}' (\mathbf{z} - \mathbf{c})}{r^2 s^2} \right). \end{aligned}$$

It holds that

$$\|\mathbf{T}'\| \leq \sum_{i=k+1}^{\infty} \left\| \frac{\Sigma_\delta}{s^2} \right\| \leq \frac{\left(\frac{\delta}{s^2} \right)^{k+1}}{1 - \frac{\delta}{s^2}}$$

then

$$\left| \pi \frac{(\mathbf{z} - \mathbf{c})^t \mathbf{T}' (\mathbf{z} - \mathbf{c})}{r^2 s^2} \right| \leq (1 + \tau)^2 \cdot n \cdot \frac{s^2 + \delta}{s^2 - \delta} \cdot \left(\frac{\delta}{s^2} \right)^{k+1} \leq \epsilon'/2.$$

Consequently, the output distribution of $\mathbf{z}$ satisfies

$$P(\mathbf{z}) \propto \left[\frac{(1 - \epsilon')^2}{(1 + \epsilon')^3}, \frac{(1 + \epsilon')^2}{(1 - \epsilon')} \right] \cdot \rho_{rs, \mathbf{c}}(\mathbf{z}).$$

By Lemma 6, we conclude that

$$R_a(P(\mathbf{z}) \mid D_{\mathcal{L}(\mathbf{B}), rs, \mathbf{c}}(\mathbf{z})) \leq \frac{(1 + \epsilon')^5}{(1 - \epsilon')^3}.$$

Expected number of repetitions. We now bound the rejection ratios Δ_1 and Δ_2 . Lemma 1 indicates that

$$\left(\frac{1-\epsilon}{1+\epsilon}\right)^{2n} \leq \Delta_1 \leq 1.$$

Given that

$$\|\mathbf{T}\| \leq \sum_{i=1}^k \left\| \frac{\Sigma_\delta}{s^2} \right\| \leq \frac{\frac{\delta}{s^2}}{1 - \frac{\delta}{s^2}} = \frac{\delta}{s^2 - \delta}$$

and

$$\|\mathbf{z} - \mathbf{c}\| \leq (1 + \tau) \sqrt{(s^2 + \delta)r} \sqrt{n/\pi}$$

we have

$$\exp\left(-\frac{(1+\tau)^2(s^2+\delta)n\delta}{(s^2-\delta)s^2}\right) \leq \exp\left(\pi \frac{(\mathbf{z} - \mathbf{c})^t \mathbf{T} (\mathbf{z} - \mathbf{c})}{r^2 s^2}\right) \leq \exp\left(\frac{(1+\tau)^2(s^2+\delta)n\delta}{(s^2-\delta)s^2}\right).$$

Since $C \geq \exp\left(\frac{(1+\tau)^2(s^2+\delta)n\delta}{(s^2-\delta)s^2}\right)$, it follows that $\frac{1}{C^2} \leq \Delta_2 \leq 1$. $\square$

Remark 6. In Algorithm 7, a moderate level of precision p can ensure practically negligible loss in security and efficiency. We showcase the example for $n = 512$: one can set the sampling width sr as per the ANTRAG^{ws} parameters while choosing $L = 4$ and $p = 2^{30}$ to make δ on the order of 2^{-18} , then $k = 1$ suffices to ensure the error ϵ' bounded by 2^{-55} .

6 Compact Gadget Trapdoors with Weak Smoothness

The instantiation of gadget trapdoors with weak smoothness is an open problem left in the conference version [ZJR⁺25]. In this section, we partially address this problem by developing a weak-smoothness variant of the compact gadget framework [YJW23] that has been used in practical lattice-based designs [YJL⁺23, EEN⁺24, BLdPP26].

In Section 6.2, we present a modified compact gadget sampler under weak smoothness and prove that the sampling procedure can be still *simulatable without knowing the trapdoor*. Interestingly, the additional χ -BDD ^{$\mathcal{O}_s$} assumption is *no longer needed* in the security proof of compact gadget signatures. Section 6.3 presents a weak smoothness variant of the HUFU signature [YJL⁺23] achieving a reduction of $> 13\%$ in signature size and $> 17\%$ in public key size.

6.1 Recall the Compact Gadget Trapdoors

The compact gadget trapdoor [YJW23] was proposed to improve the practicality of gadget-based schemes. It uses a square matrix to replace the conventional short-and-fat gadget matrix [MP12], thereby reducing the signature and public key sizes. A specialized gadget sampler—known as the semi-random sampler—was proposed to support this compact representation: it first computes the

error via deterministic decoding and then computes the preimage by Gaussian sampling.

We briefly describe the compact gadget trapdoor scheme. Let $\mathbf{P} = p \cdot \mathbf{I}_n$ and $\mathbf{Q} = q \cdot \mathbf{I}_n$ be the gadget matrices such that the modulus $Q = pq$. The public matrix is $\mathbf{A} \in \mathbb{Z}_Q^{n \times m}$ with $m > n$ and the associated trapdoor is some matrix $\mathbf{T} \in \mathbb{Z}^{m \times n}$ of small entries such that

$$\mathbf{AT} = \mathbf{P} \bmod Q.$$

Given a syndrome $\mathbf{u}$, the (approximate) preimage is computed as follows:

1. *Perturbation sampling*: Sample $\mathbf{p}$ from $D_{\mathbb{Z}^m, \sqrt{\Sigma_p}}$ where $\Sigma_p = s^2 \mathbf{I}_m - r^2 \mathbf{T} \mathbf{T}^t$
2. Compute $\mathbf{u}' = \mathbf{u} - \mathbf{Ap} \bmod Q$
3. *Semi-random gadget sampling*:
 - (a) *Deterministic error decoding*: Compute $\mathbf{e} = \mathbf{u}' \bmod p$, $\mathbf{c} = \frac{(\mathbf{u}' - \mathbf{e})}{p} \in \mathbb{Z}^n$
 - (b) *Random preimage sampling*: Sample $\mathbf{x}'$ from $D_{q\mathbb{Z}^n + \mathbf{c}, r}$
4. Output $\mathbf{x} = \mathbf{p} + \mathbf{T}\mathbf{x}' \bmod Q$

One can verify that

$$\mathbf{Ax} = \mathbf{Ap} + p \cdot \mathbf{x}' = \mathbf{u} - \mathbf{u}' + p \cdot \mathbf{c} = \mathbf{u} - \mathbf{e} \bmod Q$$

and $(\mathbf{x}, \mathbf{e})$ derives a valid signature.

Note that the distribution of compact gadget signatures is no longer Gaussian, thus the provable security needs additional treatments. It was shown in [YJW23] that for uniformly random syndromes, the sampling procedure can be simulated without knowing $\mathbf{T}$ when

$$s^2 \geq (r^2 + \eta_\epsilon(\mathbb{Z}^n)^2) \cdot (s_1(\mathbf{T})^2 + 1),$$

which gives the provable unforgeability.

6.2 Semi-random Sampler with Weak Smoothness

We present a variant of the semi-random gadget sampler under the weak smoothness condition in Algorithm 9. This reduces r from $q \cdot \eta_\epsilon(\mathbb{Z}^n)$ with negligible ϵ to $q \cdot \eta_{\epsilon'}(\mathbb{Z}^n)$ with $\epsilon' = O(1)$, giving a smaller preimage width s .

Theorem 3 shows that the modified semi-random sampler can still be simulated without the trapdoor when the syndrome $\mathbf{u}$ is uniformly random. This guarantees the provable security of compact gadget based signatures.

Theorem 3. *Let $(\mathbf{A}, \mathbf{T})$ be a matrix-approximate trapdoor pair and (r, s) such that $s^2 \geq (r^2 + \eta_\epsilon(\mathbb{Z}^n)^2) \cdot (s_1(\mathbf{T})^2 + 1)$ and $r \geq q \cdot \eta_{\epsilon'}(\mathbb{Z}^n) \geq \eta_\epsilon(\mathbb{Z}^n)$ for negligible ϵ . Then the expected number of repetitions of Algorithm 9 is at most $1 + \epsilon'$. Moreover, conditioned on successful output, the following two distributions are statistically indistinguishable:*

$$\begin{aligned} &\{(\mathbf{A}, \mathbf{x}, \mathbf{u}, \mathbf{e}) : \mathbf{u} \leftarrow U(\mathbb{Z}_Q^n), \mathbf{x} \leftarrow \text{ApproxPreSamp}^{\text{ws}}(\mathbf{A}, \mathbf{T}, \mathbf{u}, r, s), \mathbf{e} = \mathbf{u} - \mathbf{Ax} \bmod Q\} \\ &\{(\mathbf{A}, \mathbf{x}, \mathbf{u}, \mathbf{e}) : \mathbf{x} \leftarrow D_{\mathbb{Z}^m, s}, \mathbf{e} \leftarrow U(\mathbb{Z}_p^n), \mathbf{u} = \mathbf{Ax} + \mathbf{e} \bmod Q\} \end{aligned}$$

Algorithm 9: ApproxPreSamp^{ws}

Input: $(\mathbf{A}, \mathbf{T}) \in \mathbb{Z}^{n \times m} \times \mathbb{Z}^{m \times n}$, $\mathbf{u} \in \mathbb{Z}^n$ such that $\mathbf{AT} = p \cdot \mathbf{I}_n \bmod Q$,
 $r \geq \eta_\epsilon(\mathbb{Z}^n)$ and $s^2 \mathbf{I} > r^2 \mathbf{T} \mathbf{T}^t$
Output: an approximate preimage $\mathbf{x}$ such that $\mathbf{Ax} + \mathbf{e} = \mathbf{u} \bmod Q$

- 1 $\mathbf{p} \leftarrow D_{\mathbb{Z}^m, \sqrt{\Sigma_p}}$ where $\Sigma_p = s^2 \mathbf{I} - r^2 \mathbf{T} \mathbf{T}^t$
- 2 $\mathbf{u}' = \mathbf{u} - \mathbf{Ap} \bmod Q$
- 3 $\mathbf{e} = \mathbf{u}' \bmod p$ such that $\mathbf{u}' - \mathbf{e} = p \cdot \mathbf{c}$
- 4 $\mathbf{x}' \leftarrow D_{q\mathbb{Z}^n + \mathbf{c}, r}$
- 5 $\mathbf{x} = \mathbf{p} + \mathbf{T} \mathbf{x}'$
- 6 **return** $\mathbf{x}$ with probability $\Delta = \frac{\rho_{r, \mathbf{c}}(q\mathbb{Z}^n)}{\rho_r(q\mathbb{Z}^n)}$ otherwise restart

To prove Theorem 3, we need two following lemmas.

Lemma 21. Let $M = \frac{q^n \cdot \rho_r(q\mathbb{Z}^n)}{\rho_r(\mathbb{Z}^n)}$ be the expected number of repetitions. Then the following two distributions are the same.

- Distribution 1: First sample $\mathbf{u}' \leftarrow U(\mathbb{Z}_Q^n)$, $\mathbf{e} = \mathbf{u}' \bmod p$ with $\mathbf{u}' - \mathbf{e} = p \cdot \mathbf{c}$, then sample $\mathbf{x}' \leftarrow D_{q\mathbb{Z}^n + \mathbf{c}, r}$, output $(\mathbf{x}', \mathbf{u}', \mathbf{e})$ with probability $\frac{\rho_{r, \mathbf{c}}(q\mathbb{Z}^n)}{\rho_r(q\mathbb{Z}^n)}$;
- Distribution 2: First sample $\mathbf{e} \leftarrow U(\mathbb{Z}_p^n)$, then sample $\mathbf{x}' \leftarrow D_{\mathbb{Z}^n, r}$, set $\mathbf{u}' = \mathbf{e} + p \cdot \mathbf{x}' \bmod Q$, output $(\mathbf{x}', \mathbf{u}', \mathbf{e})$ with probability $\frac{1}{M}$.

Proof. The supports of two distribution are identical as follows:

$$\{(\mathbf{x}', \mathbf{u}', \mathbf{e}) \in \mathbb{Z}^n \times \mathbb{Z}_Q^n \times \mathbb{Z}_p^n \mid \mathbf{u}' = \mathbf{e} + p \cdot \mathbf{x}' \bmod Q\}.$$

The output $(\mathbf{x}', \mathbf{u}', \mathbf{e})$ of Distribution 1 has probability

$$\begin{aligned} P_1[(\mathbf{x}', \mathbf{u}', \mathbf{e})] &= \frac{1}{Q^n} \cdot \frac{\rho_r(\mathbf{x}')}{\rho_r(q\mathbb{Z}^n + \mathbf{c})} \cdot \frac{\rho_{r, \mathbf{c}}(q\mathbb{Z}^n)}{\rho_r(q\mathbb{Z}^n)} \\ &= \frac{1}{Q^n} \cdot \frac{\rho_r(\mathbf{x}')}{\rho_r(q\mathbb{Z}^n)} \\ &= \frac{1}{M} \cdot \frac{1}{p^n} \cdot \frac{\rho_r(\mathbf{x}')}{\rho_r(\mathbb{Z}^n)}. \end{aligned}$$

The output $(\mathbf{x}', \mathbf{u}', \mathbf{e})$ of Distribution 2 has probability

$$P_2[(\mathbf{x}', \mathbf{u}', \mathbf{e})] = \frac{1}{M} \cdot \frac{1}{p^n} \cdot \frac{\rho_r(\mathbf{x}')}{\rho_r(\mathbb{Z}^n)}.$$

Hence $P_1[(\mathbf{x}', \mathbf{u}', \mathbf{e})] = P_2[(\mathbf{x}', \mathbf{u}', \mathbf{e})]$. □

Lemma 22 ([YJW23, Lemma 7]). Let $r \geq \eta_\epsilon(\mathbb{Z}^n)$ for some negligible ϵ and $s^2 \geq (r^2 + \eta_\epsilon(\mathbb{Z}^n)^2) \cdot (s_1(\mathbf{T})^2 + 1)$. Then the distribution $[\mathbf{I}_m \mid \mathbf{T}] \cdot D_{\mathbb{Z}^{m+n}, \sqrt{\Sigma_p \oplus r^2 \mathbf{I}_n}}$ is statistically close to $D_{\mathbb{Z}^m, s}$.

Proof of Theorem 3. Since $r \geq q \cdot \eta_{\epsilon'}(\mathbb{Z}^n) \geq \eta_{\epsilon}(\mathbb{Z}^n)$, it follows that

$$M = \frac{q^n \cdot \rho_r(q\mathbb{Z}^n)}{\rho_r(\mathbb{Z}^n)} = \frac{\rho_{\frac{1}{r}}\left(\frac{1}{q}\mathbb{Z}^n\right)}{\rho_{\frac{1}{r}}(\mathbb{Z}^n)} \leq 1 + \epsilon'.$$

Using the notation from Algorithm 9, we introduce several intermediate hybrid distributions to prove the theorem.

Hybrid 1: Instead of sampling $\mathbf{u} \leftarrow U(\mathbb{Z}_Q^n)$, we sample $\mathbf{u}' \leftarrow U(\mathbb{Z}_Q^n)$ and $\mathbf{p} \leftarrow D_{\mathbb{Z}^m, \sqrt{\Sigma_p}}$, then compute $\mathbf{u} = \mathbf{u}' + \mathbf{A}\mathbf{p}$. In addition, the restart step becomes the step $\mathbf{u}' \leftarrow U(\mathbb{Z}_Q^n)$. We keep $(\mathbf{x}', \mathbf{x}, \mathbf{e})$ unchanged. Clearly, the real distribution and Hybrid 1 are the same.

Hybrid 2: Instead of sampling $\mathbf{u}'$, $\mathbf{x}'$ and computing $\mathbf{e}$ as in Hybrid 1, we sample $\mathbf{x}' \leftarrow D_{\mathbb{Z}^n, r}$ and $\mathbf{e} \leftarrow U(\mathbb{Z}_p^n)$, then compute $\mathbf{u}' = p \cdot \mathbf{x}' + \mathbf{e}$. All other terms $(\mathbf{p}, \mathbf{x}, \mathbf{u})$ remain unchanged. By Lemma 21, Hybrid 1 and Hybrid 2 are same.

Hybrid 3: Instead of sampling $\mathbf{p}$, $\mathbf{x}'$ and computing $\mathbf{x} = \mathbf{p} + \mathbf{T}\mathbf{x}'$ in Hybrid 2, we sample directly $\mathbf{x} \leftarrow D_{\mathbb{Z}^m, s}$ and compute $\mathbf{u} = \mathbf{A}\mathbf{x} + \mathbf{e} \bmod Q$ where $\mathbf{e} \leftarrow U(\mathbb{Z}_p^n)$ is as before. Note that $\mathbf{x}$ follows the distribution $[\mathbf{I}_m \mid \mathbf{T}] \cdot D_{\mathbb{Z}^{m+n}, \sqrt{\Sigma_p \oplus r^2 \mathbf{I}_n}}$. By Lemma 22, Hybrid 3 and Hybrid 2 are statistically close.

Now we complete the proof. $\square$

No Need for χ -BDD $^{\mathcal{O}_s}$. It is worth noting that in Theorem 3, we have already shown that the simulated signature distribution is statistically indistinguishable from the real one. This implies that the simulated syndrome $\mathbf{u}$ is uniformly distributed over $\mathbb{Z}_Q^n$. As a consequence, the security proof no longer needs the additional χ -BDD $^{\mathcal{O}_s}$ assumption introduced in Section 3.2, even though we work with a weak smoothness parameter for r .

6.3 HuFu^{ws} : A Compact Variant of HuFu

The HuFu signature scheme is a concrete signature instantiation of the compact gadget framework based on plain LWE and SIS, submitted to the NIST standardization of additional signature schemes [YJL⁺23]. Based on the modified compact gadget sampler, we present HuFu^{ws} as a weak smoothness variant of HuFu. Compared with HuFu, the signature size of HuFu^{ws} is reduced by more than 13%, and the public key is reduced by more than 17% for the NIST security levels 1, 3 and 5. Table 10 shows the parameters of HuFu^{ws}. The size reduction is less noticeable compared to the case of NTRU trapdoor based signatures (Section 3). This mainly attributes to the error term $\mathbf{e}$ whose size is independent of the preimage width s . As a result, reducing s does not permit a proportional reduction of Q , leading to a less pronounced size reduction. However, HuFu^{ws} achieves provable security without relying on the additional χ -BDD $^{\mathcal{O}_s}$ assumption.

Table 10. Parameter sets for HuFu^{ws} with $(\epsilon', \epsilon) = (2^{-1}, 2^{-49})$.

Target NIST Security Level	1	3	5
(n, m)	(680, 2160)	(960, 3104)	(1224, 3920)
Modulus Q	32768	65536	65536
Gadget parameters (p, q)	$(2^{11}, 2^4)$	$(2^{12}, 2^4)$	$(2^{12}, 2^4)$
Max. SIS norm β	29100	52088	62625
Repetitions	1.4	1.45	1.49
Public key size (in kilobytes)	847	1801	2927
Signature size (in bytes)	2079	3071	3915
Key-recovery attack: BKZ B/C/Q	442/129/117	672/196/178	876/256/232
Forgery attack: BKZ B/C/Q	438/128/116	664/194/176	881/257/233

7 Secure rANS Coding for Signature Compression

The range variant of asymmetric numeral systems (rANS) [Dud13] is an entropy coding that compresses data almost to its entropy limit with high performance. In [ETWY22], Espitau et al. proposed to use the rANS coding to compress lattice signatures and gained a 7%-14% size improvement on FALCON signatures. This technique was later adopted by two NIST additional post-quantum signature candidates HuFu [YJL⁺23] and HAETAE [CCD⁺23]. However, Saarinen demonstrated two implementation attacks on the rANS coding of HuFu and HAETAE [Saa23b, Saa23a], which results in signature forgery or buffer overflow. In this section, we present the approach to securely applying rANS for signature compression.

7.1 Countermeasure against the Bit-Flipping Attack

Saarinen discovered the bit-flipping attack against HuFu [Saa23b], i.e. an adversary can flip some bits in HuFu signature and get a new valid signature for the same message. In fact, the bit-flipping attack would generally affect all rANS encoded signatures, e.g. HAETAE. The root cause of the bit-flipping attack lies in the non-uniqueness of coding when the encoders start from different initial states [Dud09].

To overcome this issue, we exploit the unique decoding property of rANS: for any symbol and state, there is only one previous state that can be decoded to them [Dud13, Gie14]. This implies that given a fixed final state, there is only one rANS coding that can be decoded to a specific message. For this, it suffices to choose a constant number as the initial state of the encoder and to only accept the decoded message if the final state of the decoder matches that number, then each message has a unique rANS coding. In practice, most of the rANS implementations [Bon22, Joh22, Col22] have already fixed the initial state for the

Algorithm 10: rANSEncode

Input: The radix b , the state interval I , the initial state for the encoding $x_{init} \in I$, the width of the packed state w , the state intervals of symbols I_{sym} , the symbols to encode $(sym_1, \dots, sym_n)$
Output: A rANS coding $(str_{j-w+1}, \dots, str_j, str_{j+1}, \dots, str_{-1})$

```

1  $x \leftarrow x_{init}$                                 /* Fix the initial state for encoding */
2  $j \leftarrow -1$ 
3 for  $i = n, \dots, 1$  do                          /* Encode symbols backwards */
4   while  $x \notin I_{sym_i}$  do                      /* Normalize the state */
5      $str_j \leftarrow x \bmod b$ 
6      $j \leftarrow j - 1$                           /* Emit data backwards */
7      $x \leftarrow \lfloor x/b \rfloor$ 
8   end while
9    $x \leftarrow C(sym_i, x)$                       /* Encode a symbol */
10 end for
11  $(str_{j-w+1}, \dots, str_j) \leftarrow \text{PackState}(x)$ 
12 return  $(str_{j-w+1}, \dots, str_j, str_{j+1}, \dots, str_{-1})$ 

```

encoding as shown in Algorithm 10, but they do not verify the final state of the decoding, e.g. HuFu and HAETAE. Hence we modify these decoders to check if their final state matches the encoder’s initial state. We also verify that the decoder’s initial state is within a valid range and the input coding is entirely used up. The modifications are marked as blue in Algorithm 11. This guarantees that the messages and codings are bijective, then defeats the bit-flipping attack. These modifications can be easily integrated into existing rANS implementations and only have minor impact on the performance.

7.2 Countermeasure against the Length Modification Attack

Since rANS is a variable length coding, the compressed data has different lengths for different messages. Therefore, both HuFu and HAETAE add fields to their signatures to indicate the length of the compressed signatures. However, Saarinen pointed out that an adversary can tamper with the length field to trigger buffer overflows [Saa23b, Saa23a]. To avoid this risk, we propose to pad the compressed signature to a fixed length L . For most implementations that always emits full bytes and fills the buffer backwards, we could use byte padding:

$$0x00^{L-|str|-1} \parallel 0x80 \parallel str,$$

which resembles ISO/IEC 7816-4 [CP13].

8 Conclusion and Future Work

In this paper, we have demonstrated the feasibility of the GPV framework with weak smoothness condition through several positive results including theoretical

Algorithm 11: rANSDecode

Input: The radix b , the state interval I , the initial state for the encoding $x_{init} \in I$, the width of the packed state w , the number of symbols n , a rANS coding $(str_1, \dots, str_N)$

Output: The decoded symbols $(sym_1, \dots, sym_n)$ or $\perp$

```

1 if  $w > N$  then return  $\perp$  /* The coding is too short */
2  $x \leftarrow \text{UnpackState}(str_1, \dots, str_w)$  /* Read the initial state */
3  $j \leftarrow w + 1$ 
4 if  $x \notin I$  then return  $\perp$  /* The initial state is out of range */
5 for  $i = 1, \dots, n$  do
6    $(sym_i, x) \leftarrow D(x)$  /* Decode a symbol */
7   while  $x \notin I$  do /* Normalize the state */
8     if  $y > N$  then return  $\perp$  /* The coding is too short */
9      $x \leftarrow xb + str_j$ 
10     $j \leftarrow j + 1$ 
11   end while
12 end for
13 if  $x \neq x_{init}$  then return  $\perp$  /* Not match the initial state */
14 if  $j \neq N + 1$  then return  $\perp$  /* The coding has redundant parts */
15 return  $(sym_1, \dots, sym_n)$ 

```

security proof, compatible Gaussian samplers and practical signature instantiations. This introduces new tradeoffs among security, efficiency and simplicity in lattice-based cryptography designs. Our work sets an initial grounding for the use of weak smoothness and there are still many related problems to be studied.

Application to Hawk signatures. Hawk [DPPW22] is a hash-and-sign signature scheme based on a module variant of lattice isomorphism problem [DvW22]. Hawk offers appealing efficiency comparable to Falcon, along with a much simpler implementation entirely over integers. In contrast to what is presented in Section 3, the *uniform syndrome closeness* in Hawk is different as follows:

$$\Delta(U(\mathbb{Z}_2^n), \mathbf{u} = D_{\mathbb{Z}^n, s} \bmod 2) = O(\epsilon).$$

One may need other assumptions or different rejection sampling to give a formal security proof, which we leave as an interesting future work.

Application to gadget trapdoors. Lattice gadget trapdoors are a crucial building block for the constructions of advanced lattice cryptosystems. While we have presented a construction of compact gadgets under the weak smoothness condition in Section 6, adapting general gadget trapdoors to weak smoothness requires to overcome two main technical issues. First, our modified Peikert sampler (Algorithm 3) uses a scaling factor to ensure the Σ_3 -condition, which makes the perturbation is no longer an integer vector. However, the integrality of perturbations seems necessary in gadget trapdoors. Additionally, the adaption

of gadget sampling needs the calculation of a high-dimensional theta function $\rho_{s,c}(\mathcal{L}_{\mathbf{g}})$ over the gadget lattice $\mathcal{L}_{\mathbf{g}}$, which seems complicated.

Precise estimate of smoothing parameters. In GPV schemes, the achieved preimage Gaussian widths correspond to an upper bound of the smoothing parameter of q -ary lattices. However, the gap between the upper bound and the actual value could be unclear. A precise estimate of cryptographic lattice smoothing parameters would give a better understanding of the uniform syndrome closeness and Gaussian sampling closeness.

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